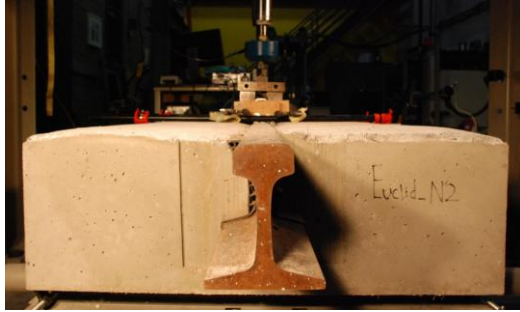


Large Scale Testing and Analytical Modeling of FRC & UHPC



Devansh Patel, Ph.D. Student, M.S., EIT

School of Sustainable Engineering and the Built Environment

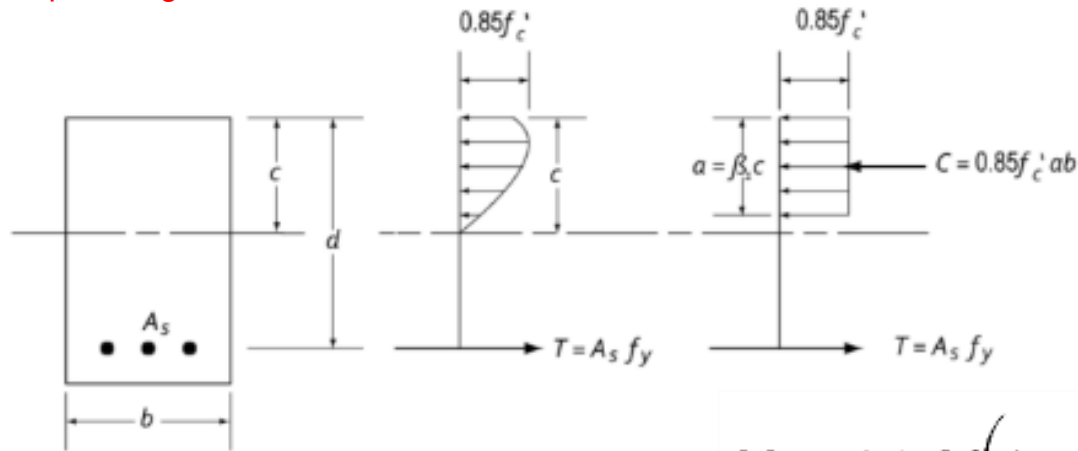
Arizona State University, Tempe, AZ 85287-5306

Outline

- Introduction
- Valley Metro Testing (Light Rail for Phoenix City)
- Precast Tunnel Lining Testing
- Slab Panel Testing
- Analytical Modeling + Inverse Analysis (FRC, UHPC, Hybrid RC)
- Case Study of Ductility of Hybrid UHPC
- Machine Learning
- Software Tool Available

Concepts for the structural design are more than 120 years old

- Structural Design of reinforced concrete has been practically unchanged in 120+ years
- Guidelines developed in 1907, modified in 1930s, and codified in 1940s.
- Approximately 60-70% of concrete's volume that is subjected to tensile stress is still ignored and thus a waste of resource, providing no contribution other than providing a cover for the steel rebar.



Rectangular Stress Block
Charles S. Whitney, 1930s

$$M_u = \phi A_s f_y d \left(1 - 0.59 \frac{\rho f_y}{f'_c} \right)$$

Formulas.

Position of neutral axis,

$$k = \sqrt{2n \left(p + p' \frac{d'}{d} \right) + [n(p + p')]^2 - n(p + p')} \quad (26)$$

Position of resultant of compressive stress, $C + C'$,

$$x = \frac{k + \frac{dC'}{dC}}{1 + \frac{C'}{G}}; \text{ in which } \frac{C'}{C} = \frac{2p'n \left(k + \frac{d'}{d} \right)}{k^2} \quad (27)$$

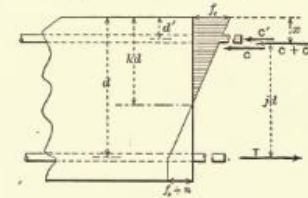


FIG. 67.

Arm of resisting couple,

$$j = \left(1 - \frac{x}{d} \right) \quad (28)$$

Moment of resistance,

$$M_s = f_s p j \cdot b d^2 \quad (29)$$

$$M_s = \frac{1}{2} f_c k (1 - \frac{1}{2} k) b d^2 + f'_s p' b d (d - d') \quad (30)$$

Fibre stresses,

$$f_s = \frac{M + j d}{A} \quad (31)$$

$$f_c = \frac{k}{n(1-k)} \cdot f_s \quad (32)$$

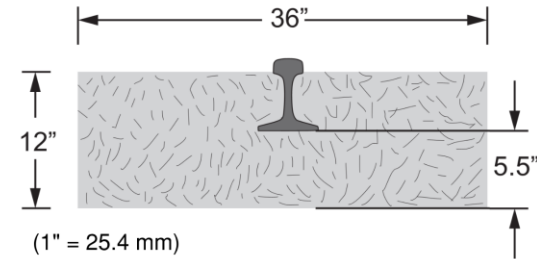
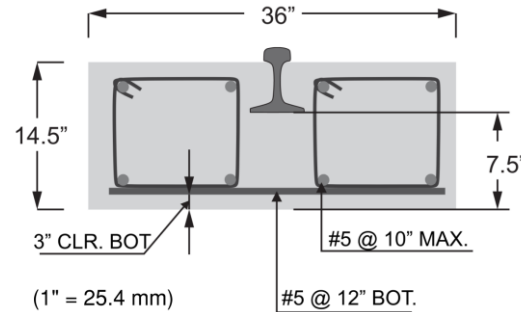
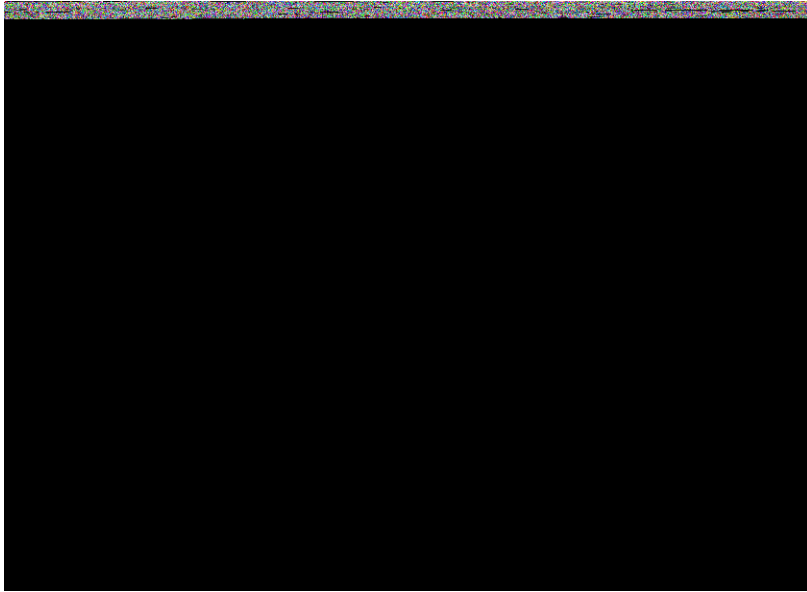
$$f'_s = \frac{n \left(k - \frac{d'}{d} \right)}{k} \cdot f_c = \frac{k d - d'}{d - k d} \cdot f_s \quad (33)$$

Large Scale Test : Pheonix Valley Metro (Light Rail)

Northwest Phase 2 project with 12,000 feet (3.5 km) of the 8' (2.5 m) wide track slab and South-Central Phase with 55,000 feet (17 km) of the 8' wide (2.5 m) track slab.

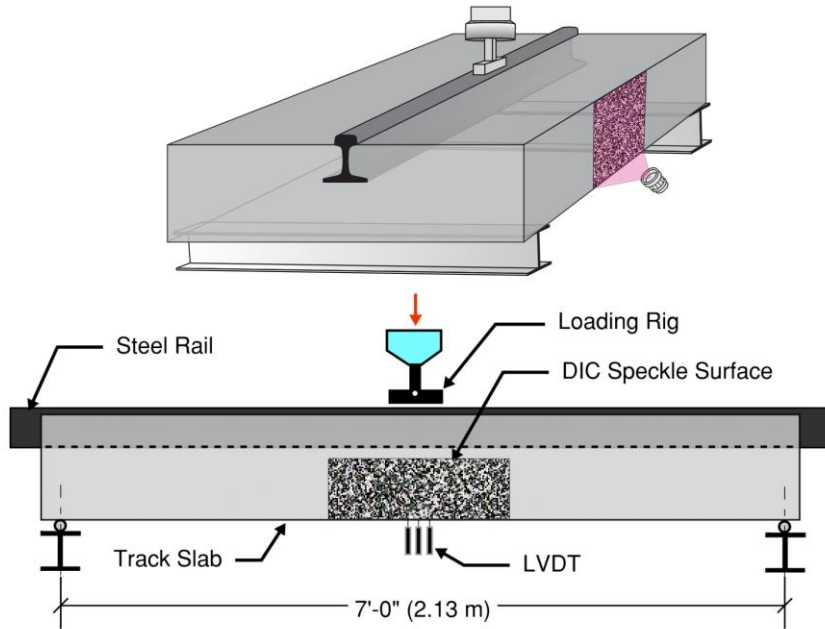


Large Scale Test : Pheonix Valley Metro (Light Rail)



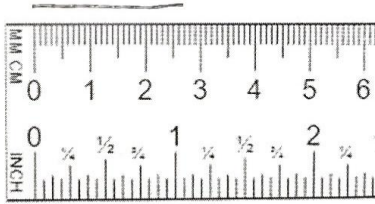
Large Scale Test : Pheonix Valley Metro (Light Rail)

Monotonic and Fatigue Testing



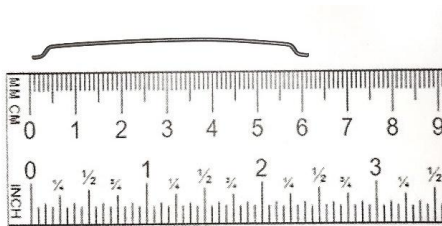
Large Scale Test : Pheonix Valley Metro (Light Rail)

Monotonic Test Results



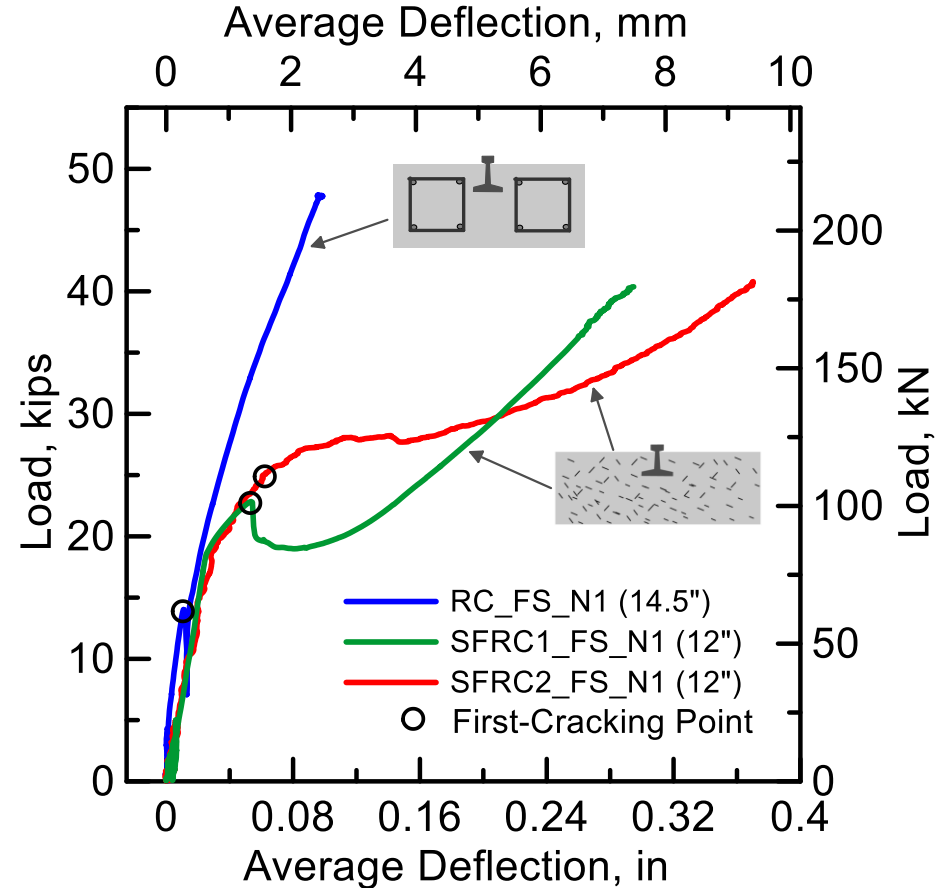
Type 1

Twisted fibers (2.5 cm),
25 pcy (15 kg/m³), 0.2% V_f



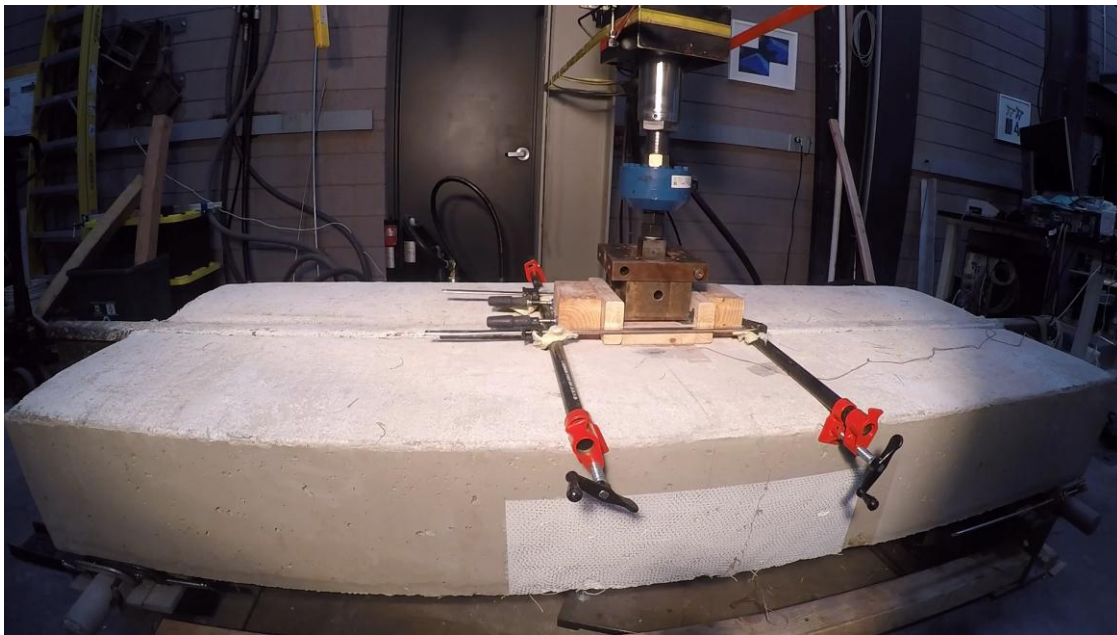
Type 2

Hooked Steel fibers (6 cm),
65 pcy (39 kg/m³), 0.5% V_f



Large Scale Test : Pheonix Valley Metro (Light Rail)

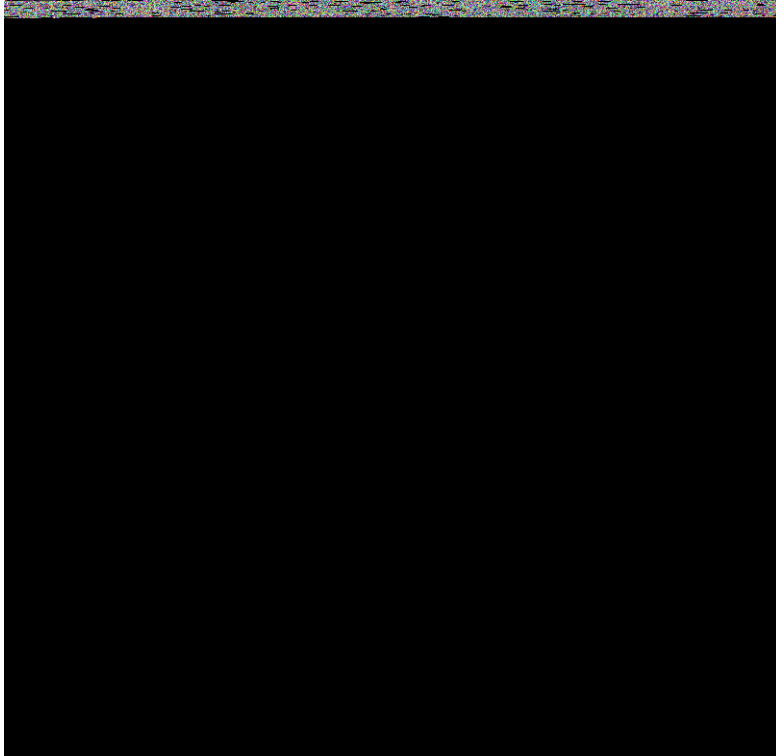
Fatigue Test Results



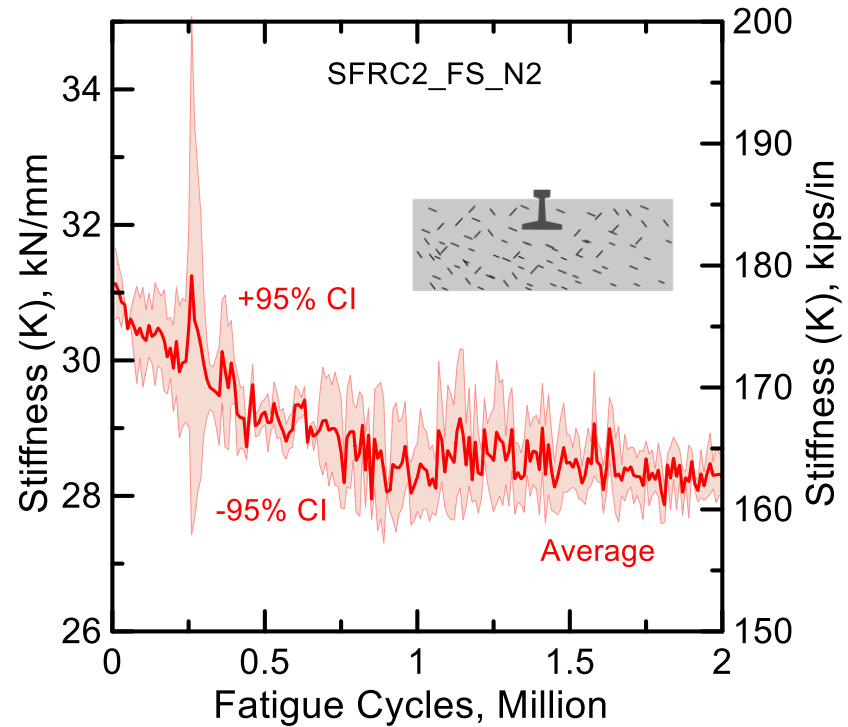
-Loading : 17.5 kips (78 kN) = **2x Service Loading**

-Data Acquisition of 100 cycles per 10,000 cycles, 2.0 million cycles, 8 days at 4 Hz

Large Scale Test : Pheonix Valley Metro (Light Rail)

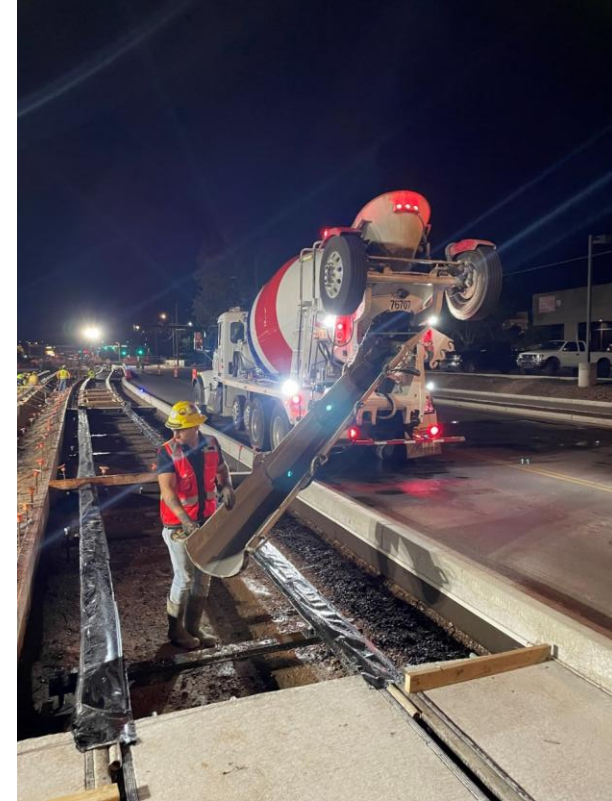


A service life of 30 years



9.5% Reduction at 2x service loading

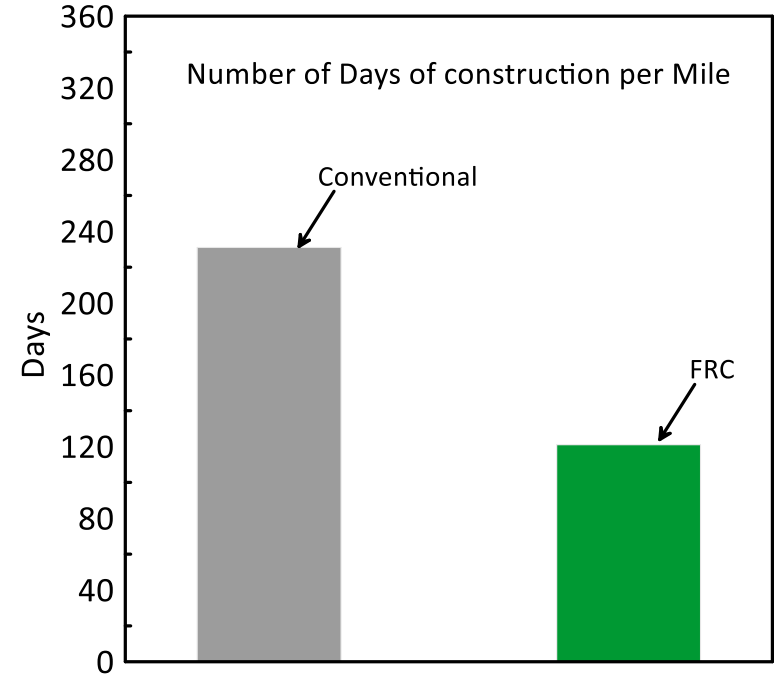
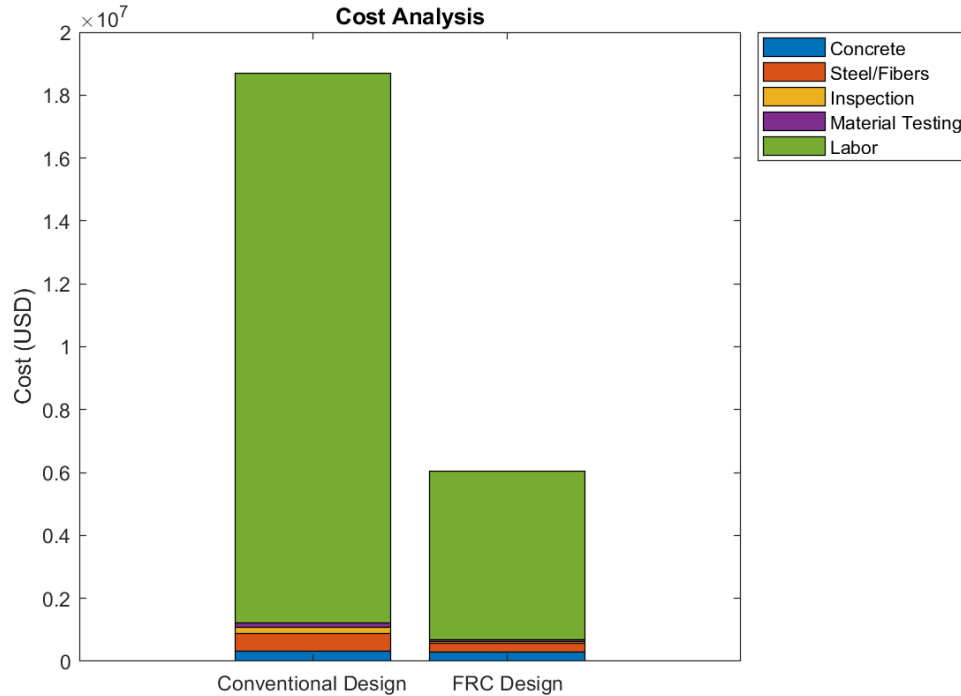
Large Scale Test : Pheonix Valley Metro (Light Rail)



Large Scale Test : Pheonix Valley Metro (Light Rail)

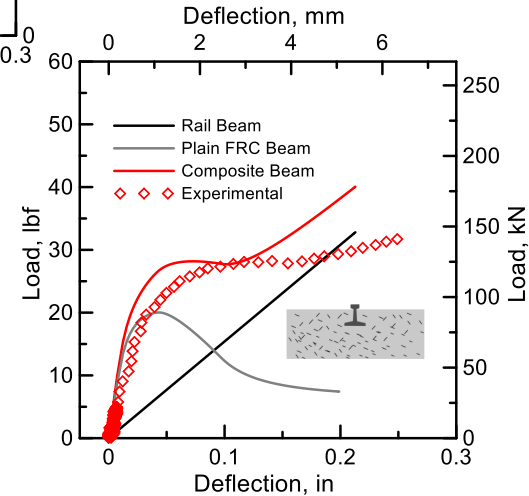
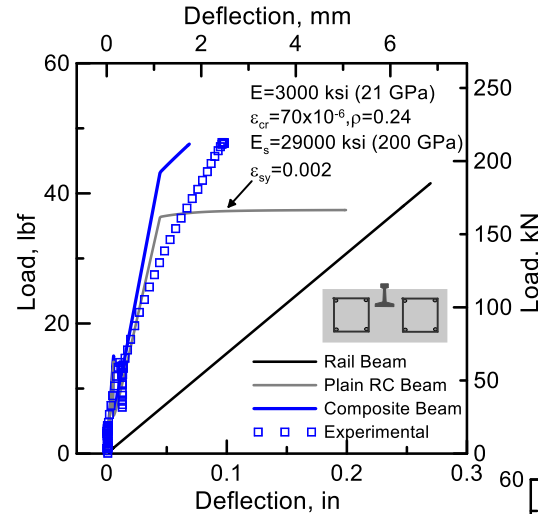
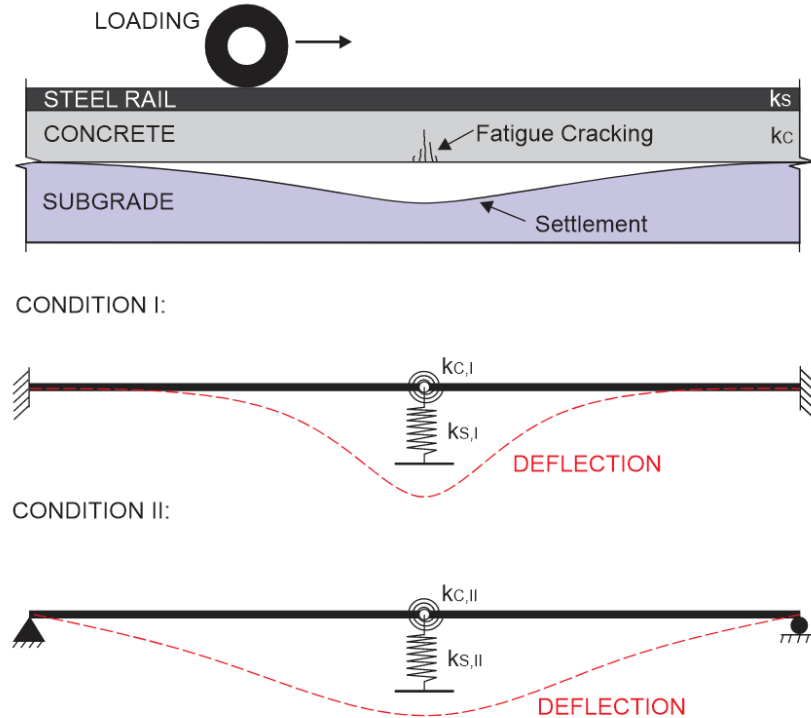
Unit of measure 16' wide x 1 mile	Original Design approach	Proposed Design Approach
Length, depth	5280' x 14.5" thick	5280' x 12" thick
Volume of Concrete used in yd ³	3651 cubic yards	3129 cubic yards
Steel Rebar used	184,000 lbs #4, 298000 lbs #5	
Steel Fiber 60 lbs per cubic yard		203,000 lbs steel fibers, 2"
Costs		
Concrete @ \$90 per yard	3651(90)= \$328,300	3129(\$90)= \$281,610
Steel cost Rebar vs. Fiber (\$90/60 lbs)	#4 \$211,600+ #5\$342,700= \$554,300	3129(\$90)= \$281,610
Inspection	\$193,527	\$75,400
Materials Testing	\$136,000	\$53,000
Cost steel rebar	\$554,300	
Cost steel fibers		\$281,610
Materials only Cost Savings/mile		-\$527,193
1 track ft cost in place	\$3309	\$1013
Total cost per mile	\$17,471,000	\$5,348,000
Total Cost savings/mile		-\$12,122,800
Project Duration for 1 mile		
A 480' section scheduled (1 labor day is equivalent to 10 workers)	21 days	11 days
Differential per section	-10	
A mile of track	231 days	121 days
Differential per mile	110 days	
Labor Costs	\$3309 per track ft labor	\$1013 per track ft labor
Total cost per mile	\$17,471,520	\$5,348,640

Large Scale Test : Pheonix Valley Metro (Light Rail)



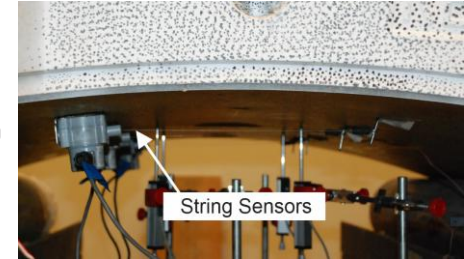
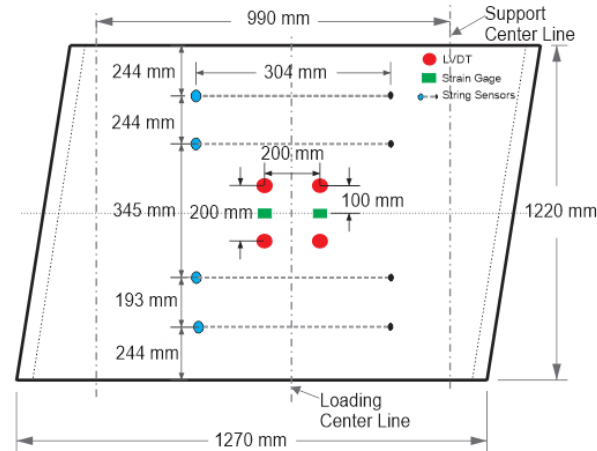
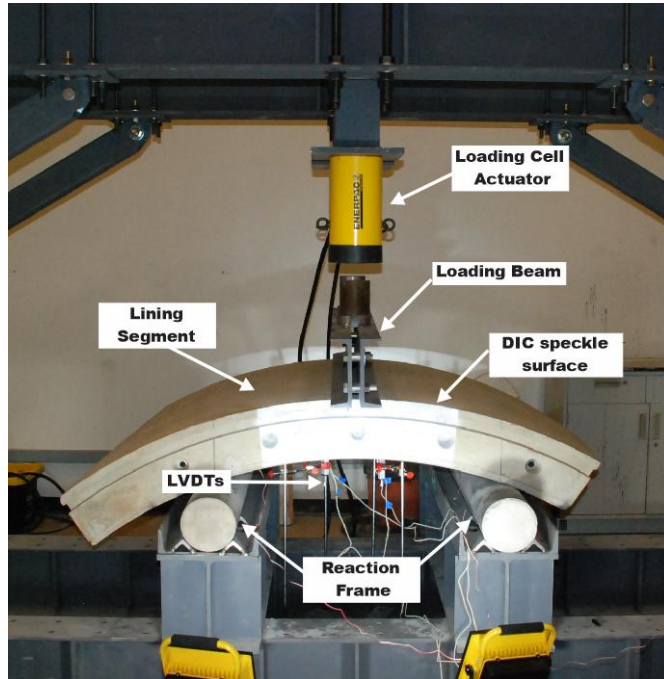
Large Scale Test : Pheonix Valley Metro (Light Rail)

Analytical Modeling



Large Scale Test : Tunnel Lining

Aim: Replace Steel Fibers with PP Fibers for Waste Water Tunnels in Los Angeles

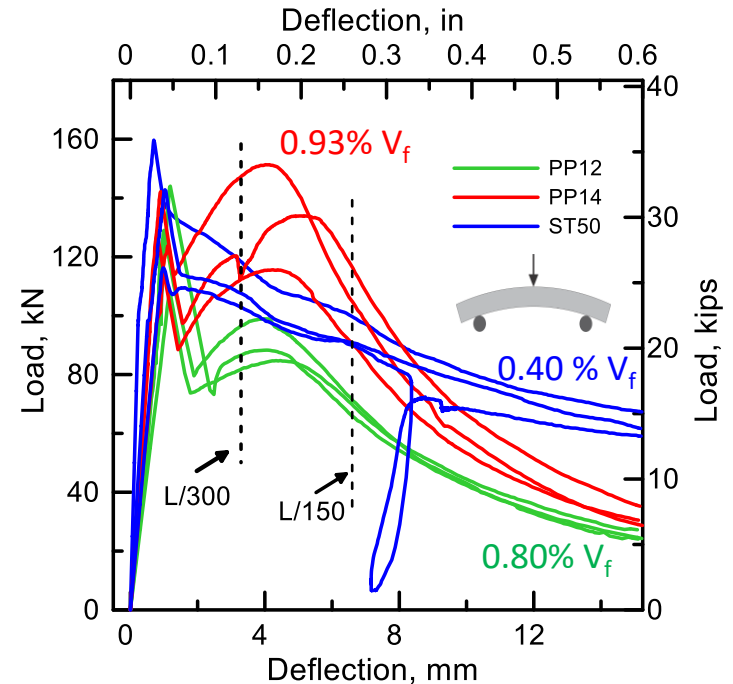
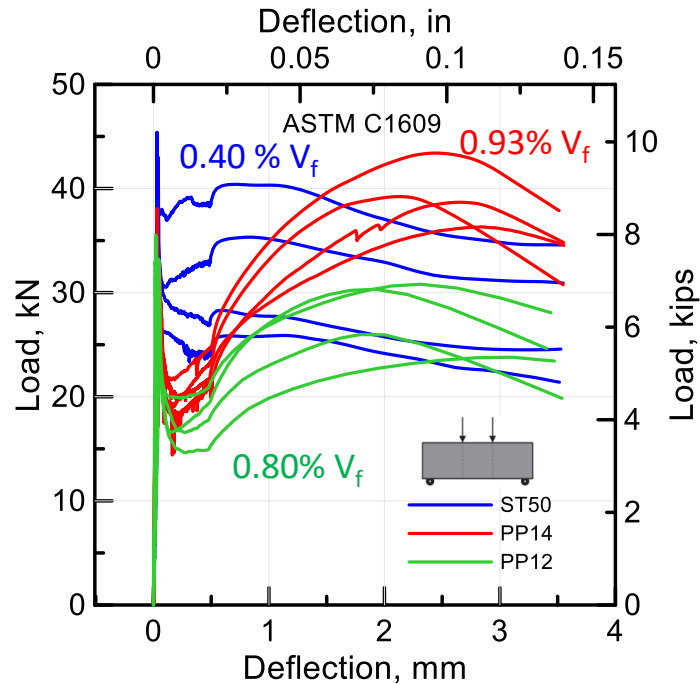
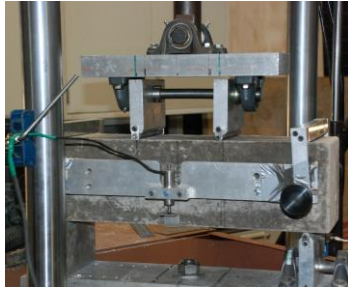


[Mechanical response of precast tunnel segments with steel and synthetic macro-fibers](#)

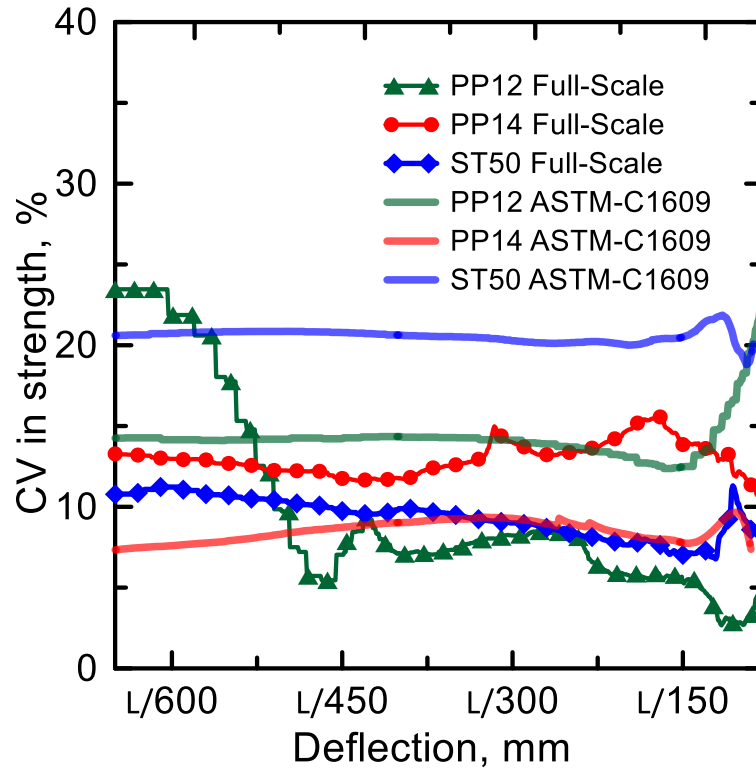
D Patel, C Pleesudjai, V Bui, P Pridemore, S Schaef... - Cement and Concrete Composites, 2023

Large Scale Test : Tunnel Lining

Flexure Results



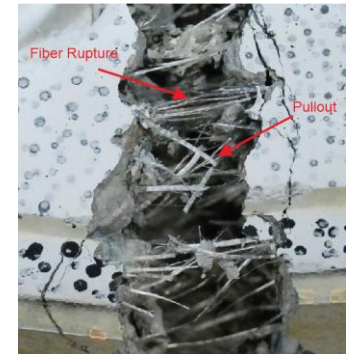
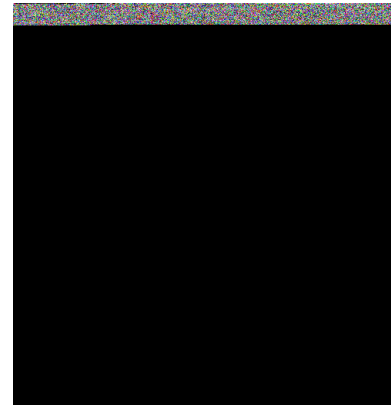
Large Scale Test : Tunnel Lining



Small Scale Specimens has larger Variation in flexure response compared to larger tunnel segments.

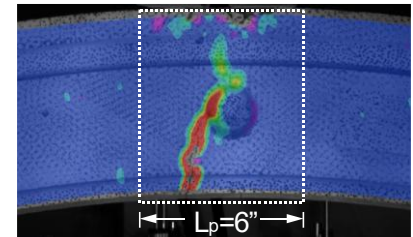
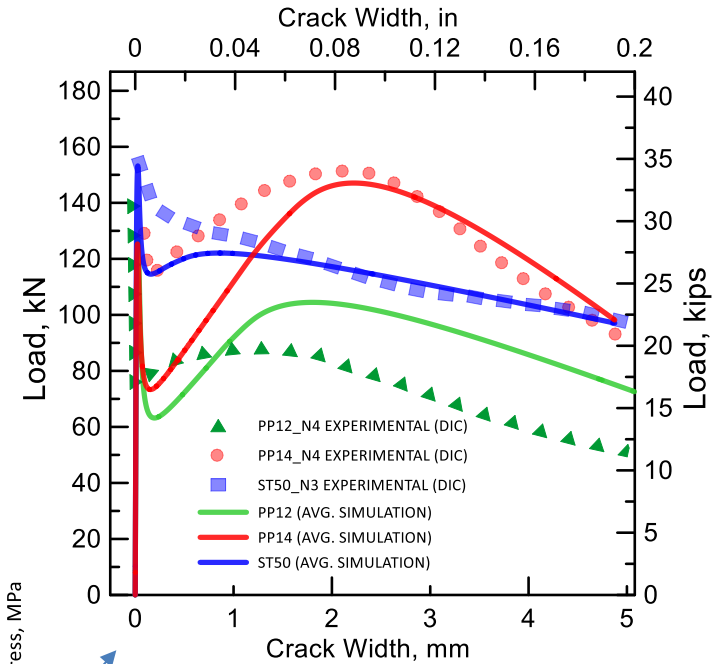
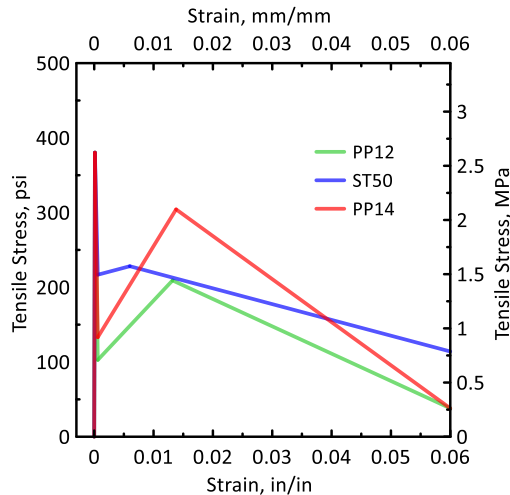
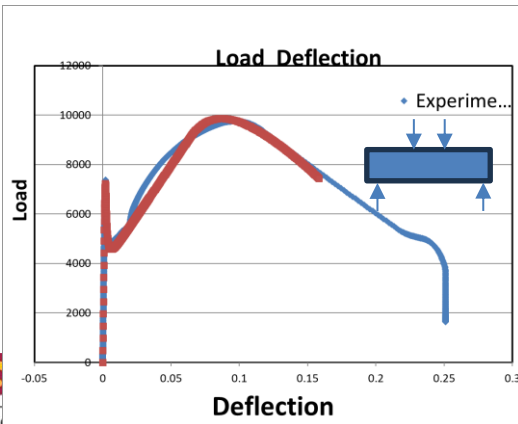
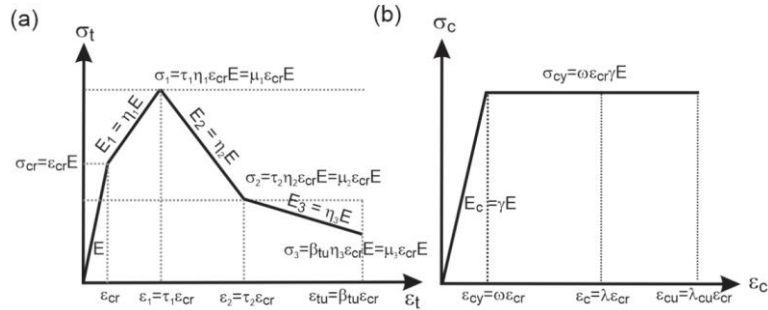
Averaging effect of fiber distribution due to larger cross-section.

10x CS leads results in 50% reduction of CV.



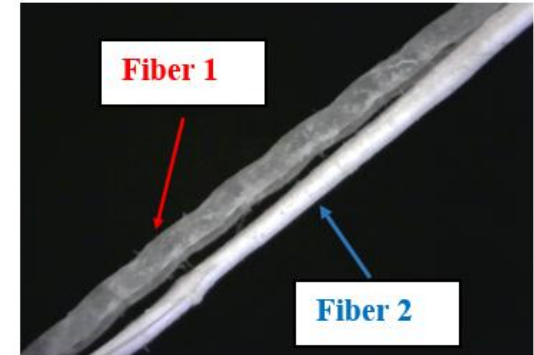
Large Scale Test : Tunnel Lining

Analytical Modeling (Quad-Linear Model)



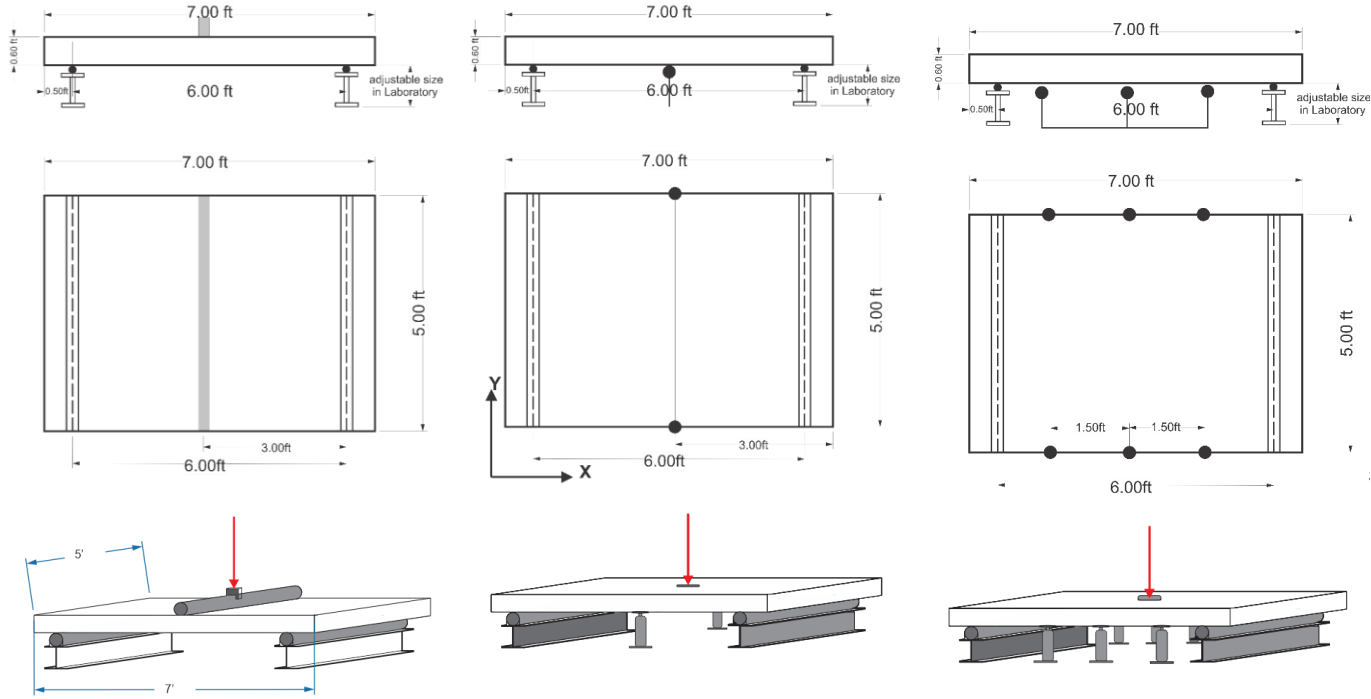
Large Scale Test : Hybrid Slab Tests

Runoff Water Tanks



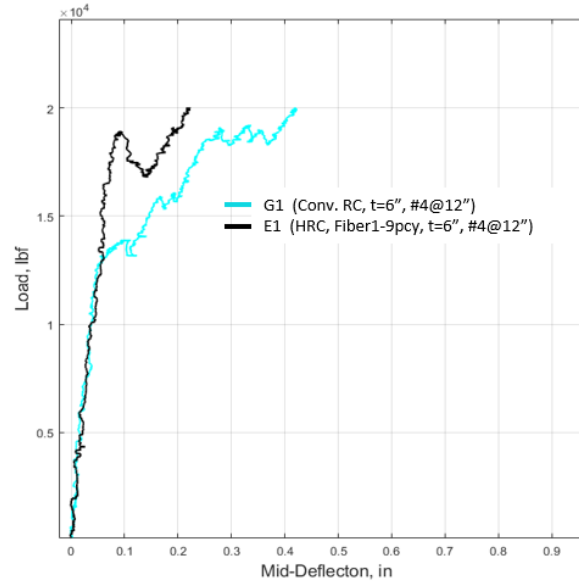
Large Scale Test : Hybrid Slab Tests

Full-Scale panel testing (3 Different Boundary Conditions)



Large Scale Test : Hybrid Slab Tests

PP Fibers increase service level performance



Large Scale Test : Hybrid Slab Tests

Plain Concrete Panel failing with punching shear

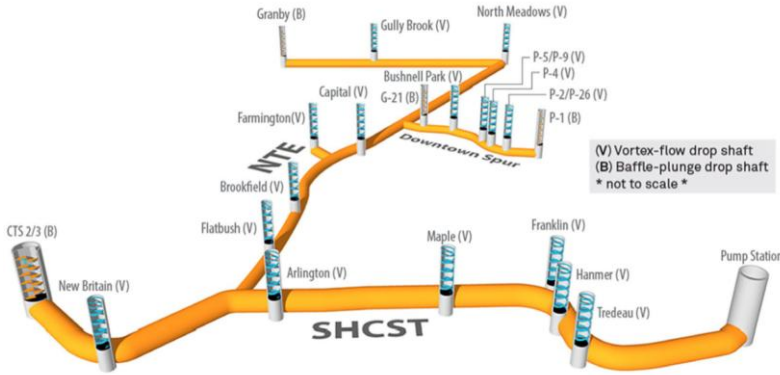
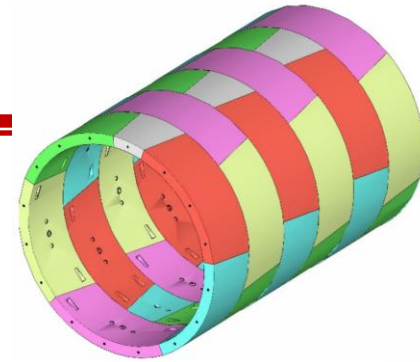


Back-calculation of mechanical properties of fiber-reinforced concrete in tunnel lining segments

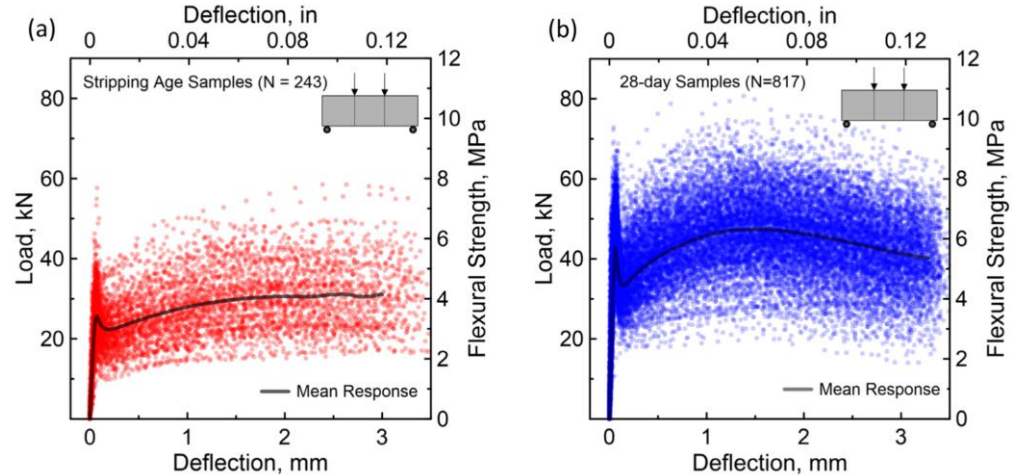
Journal | Structural Concrete. 2025 (International Federation for Structural Concrete)

<https://doi.org/10.1002/suco.70052>

Devansh Patel, Chidchanok Pleesudjai, Mehdi Bakshi, Verya Nasri, Barzin Mobasher

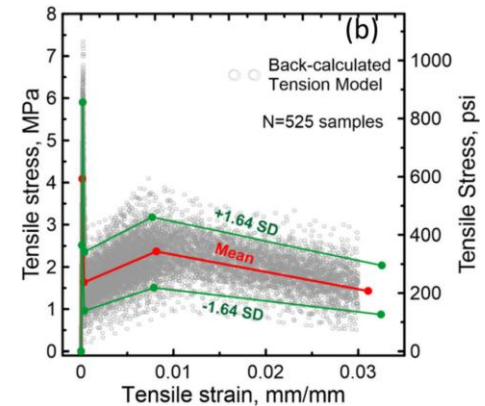
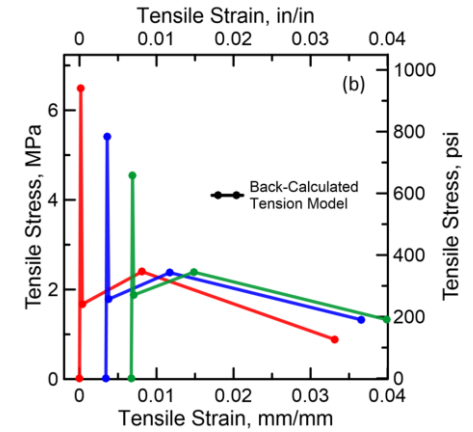
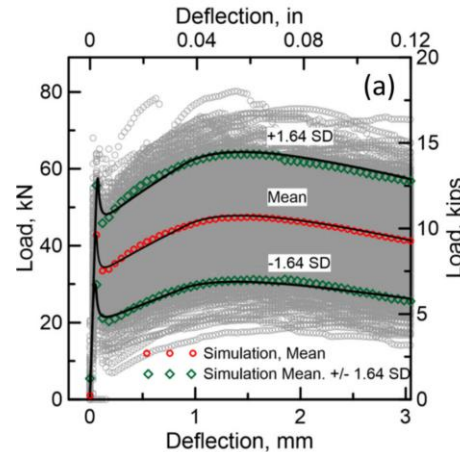
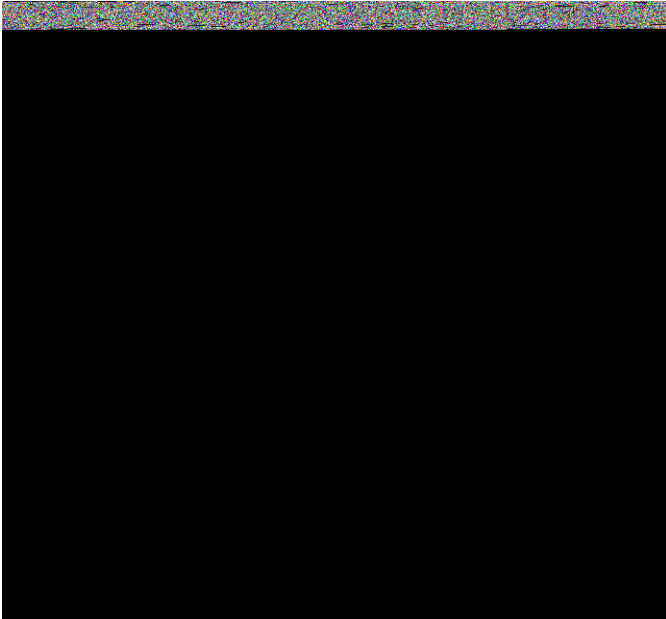


SHCST and North Tunnel Extension (NTE)

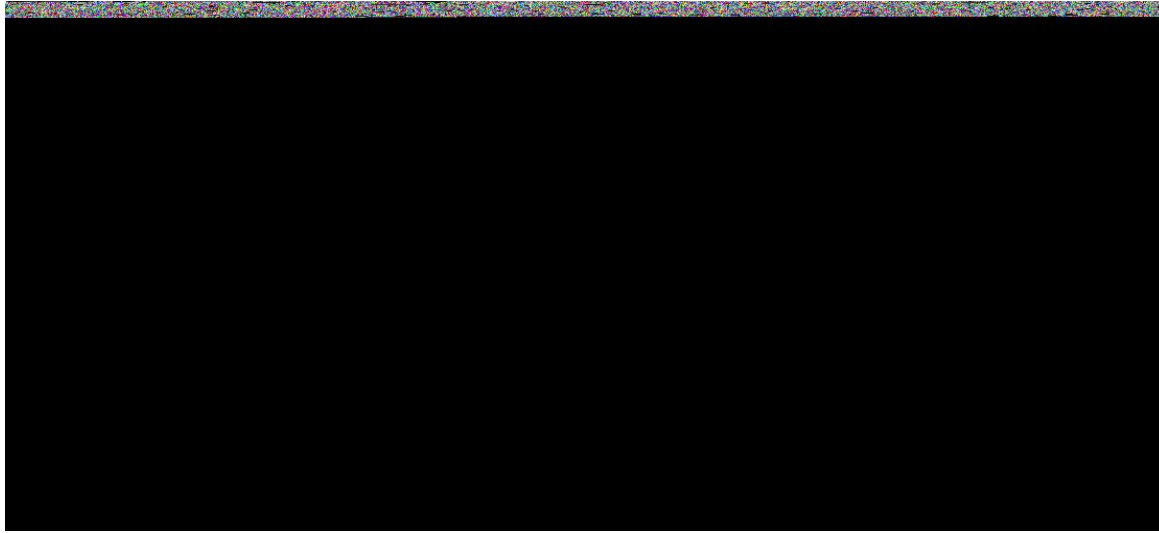


Scatter plot of four-point bending test (4PBT) results tested according to ASTM C1609; a) Scatter plot with mean response of stripping-age specimens; b) Scatter plot with mean response of 28-day specimens

-First Crack Strength (MOR) vs. Residual Strength



Average tensile stress calculation for simplified design



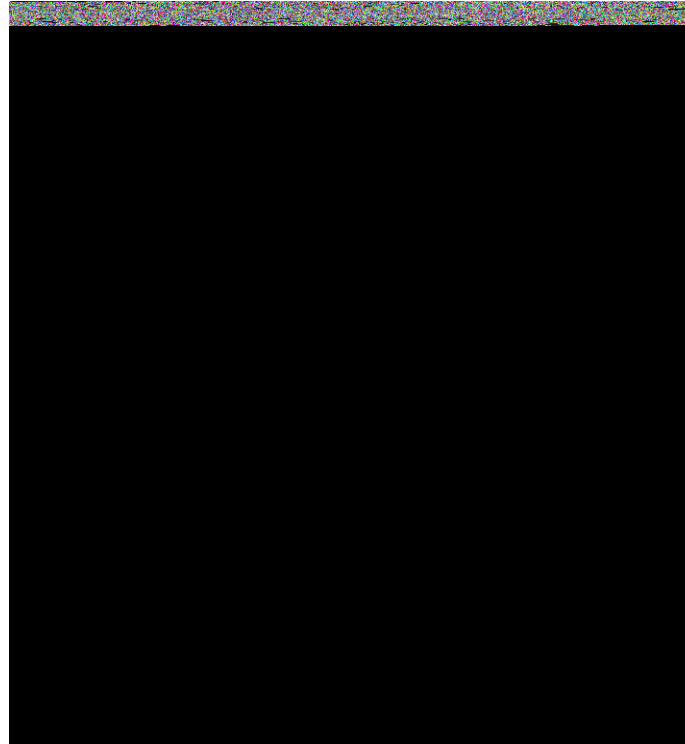
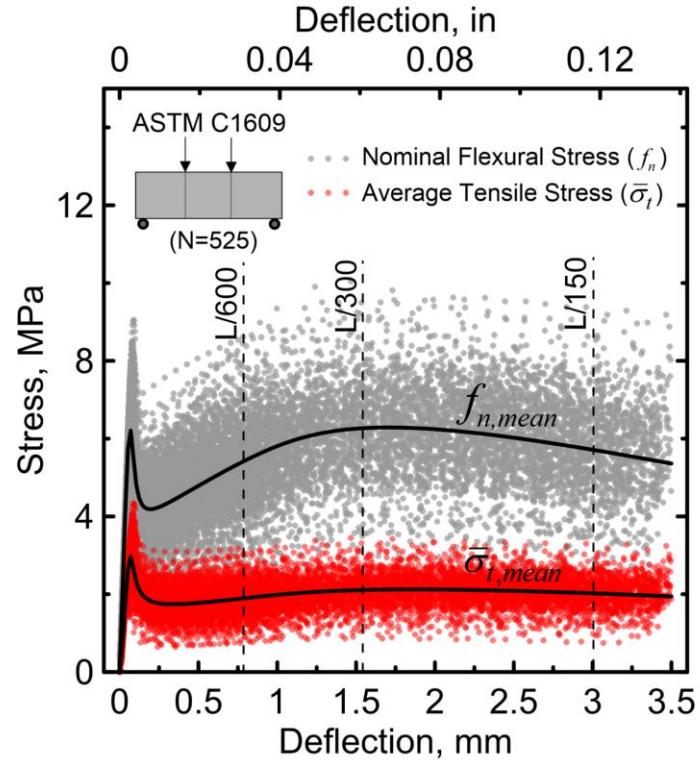
Total Force in T/C

$$F_C = F_T = \sum_0^{(1-k)h} F_t(\varepsilon) = b \int_0^{(1-k)h} \sigma(\varepsilon) dy = bh \int_0^{\beta_i} \sigma_t(\beta_i) \left(\frac{(1-k)}{\beta_i} \right) d\beta_i$$

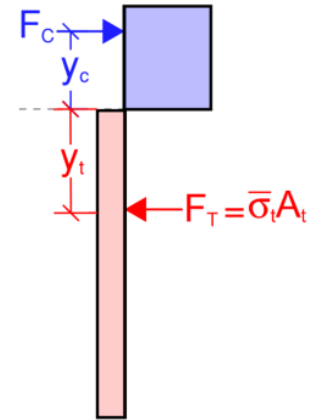
Avg. Force in T/C

$$\bar{\sigma}_t = \frac{F_T}{A_t} = \frac{F_T}{(1-k)hb}$$

Average tensile stress calculation for simplified design

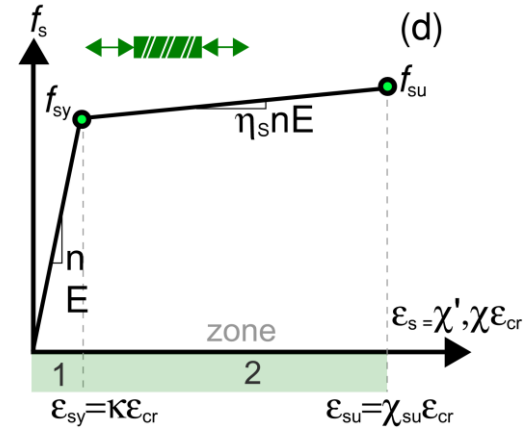
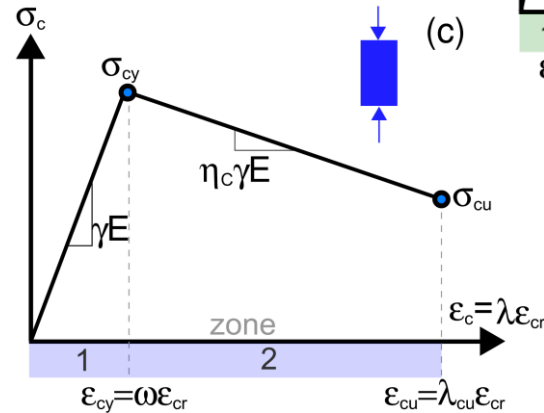
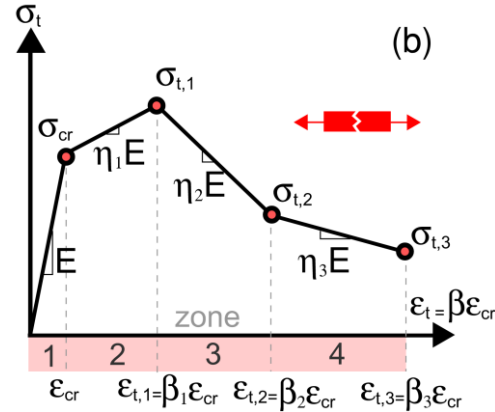
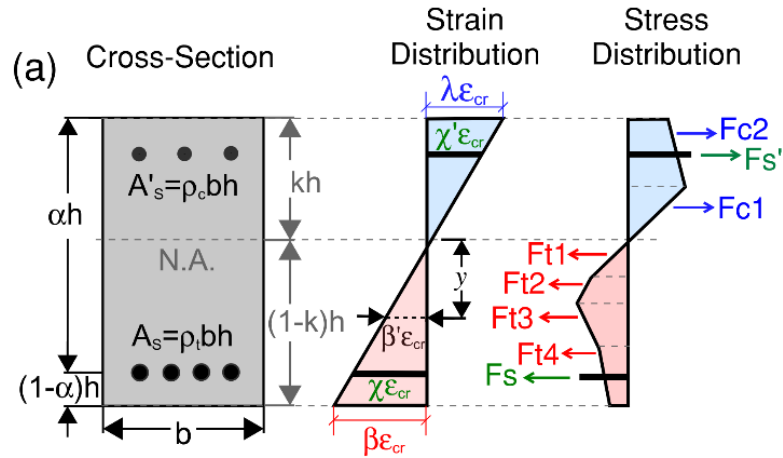


Simplified Stress Distribution



Closed-Form Analytical Modeling

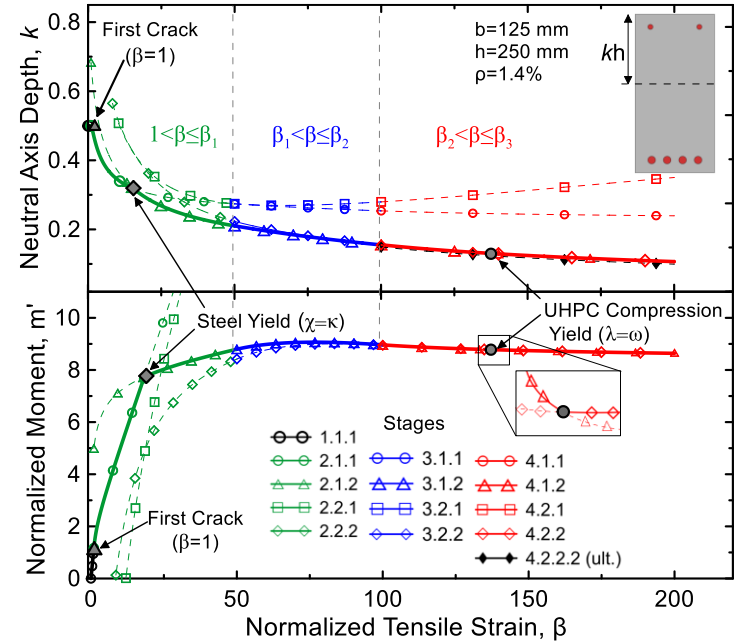
- Flexure (UHPC/FRC/HRC)



Analytical Modeling

$$F(\beta) = b \int_0^h \sigma(\beta') dy = F_{CT} + F_{CC} + F_{ST} + F_{SC} = 0$$

$$M(\beta) = b \int_0^h y \sigma(\beta') dy \quad \varphi = \frac{d\theta}{dx} = \frac{\beta \varepsilon_{cr}}{(1-k)h}$$



Closed-Form Equations

Stage	k	m'
1.1.1 $0 < \beta < 1$	$k_{111} = \frac{\sqrt{A_1^2 n^2 + (A_2 + A_3)n + \gamma - 1} + A_1 n}{\gamma - 1}$	$m'_{111} = -\frac{6\beta(C_1 k^3 + C_2 k^2 + C_3 k + C_4 n)}{(k-1)}$
2.1.1 $1 < \beta < \beta_1$	$k_{211} = \frac{\sqrt{(A_4 n \alpha + A_5(-1 + \beta)^2 \eta_1 + n A_6 \beta^2 + A_7) \beta^2 + A_8 \beta^2 + A_9}}{(\gamma - \eta_1) \beta^2 + A_9}$	$m'_{211} = -\frac{6((C_3 k^2 + C_6 k + C_7) \beta^3 + C_8)}{\beta^2 (k-1)}$

$$A_1 = (\rho_c - \rho_t)$$

$$A_2 = ((2\gamma - 2)\alpha - 2\gamma)\rho_c$$

$$A_3 = 2(1 + (\gamma - 1)\alpha)\rho_t$$

$$A_4 = -2(\rho_c - \rho_t)(-(-1 + \beta)^2 \eta_1 + \beta^2 \gamma - 2\beta + 1)$$

$$A_5 = 2\left(n\rho_t + \frac{\gamma}{2}\right)$$

$$A_6 = (n\rho_c^2 + (2n\rho_t + 2\gamma)\rho_c + n\rho_t^2)$$

$$A_7 = (4n\rho_t + 2\gamma)\beta - 2n\rho_t - \gamma$$

$$A_8 = (-\eta_1 + (-\rho_c - \rho_t)n)$$

$$A_9 = (2\eta_1 - 2)\beta - \eta_1 + 1$$

$$C_1 = \frac{1}{3}(\gamma - 1)$$

$$C_2 = (1 + (\rho_t - \rho_c)n)$$

$$C_3 = (-1 + ((-2\alpha + 2)\rho_c - 2\rho_t \alpha)n)$$

$$C_4 = \frac{1}{3} + \left(-(\alpha - 1)^2 \rho_c + \rho_t \alpha^2\right)$$

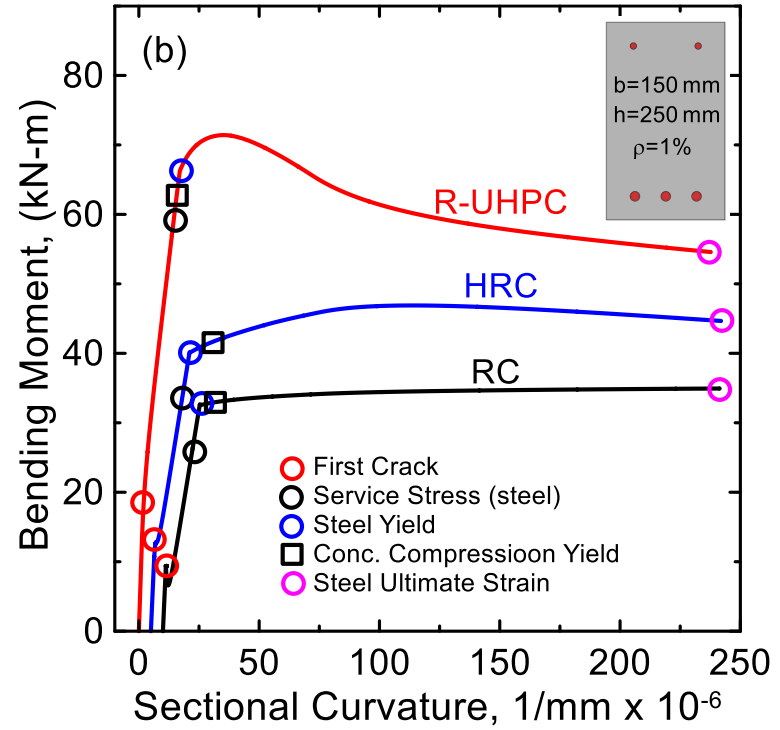
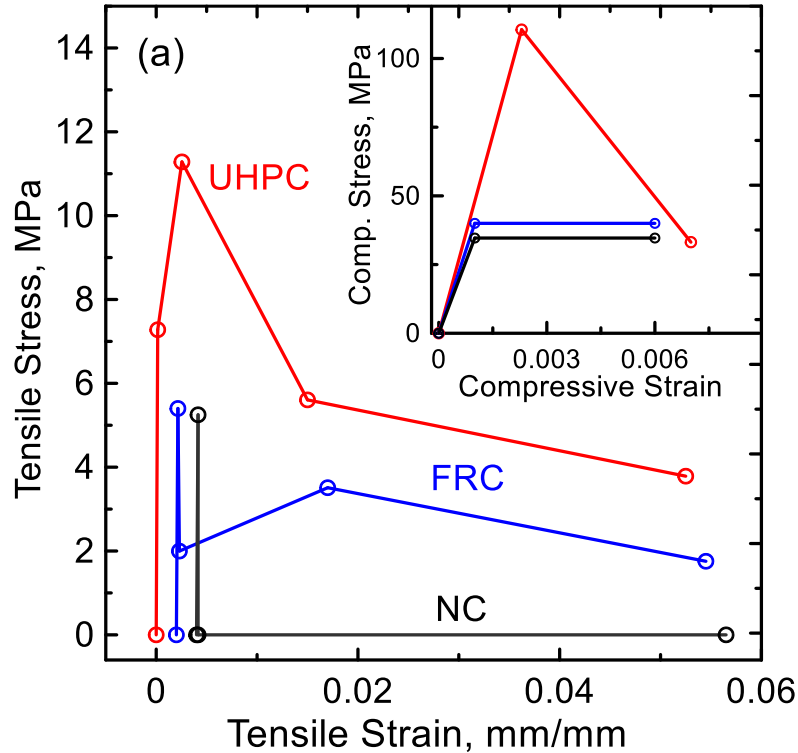
$$C_5 = \left(-\frac{\eta_1}{3} + \frac{\gamma}{3}\right)k^3 + (\eta_1 + (\rho_c + \rho_t)n)$$

$$C_6 = (-\eta_1 + ((2\alpha - 2)\rho_c - 2\rho_t \alpha)n)$$

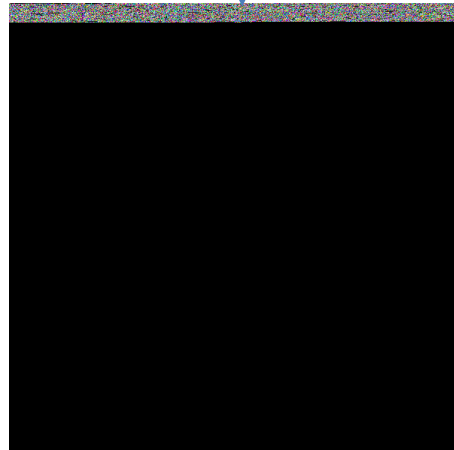
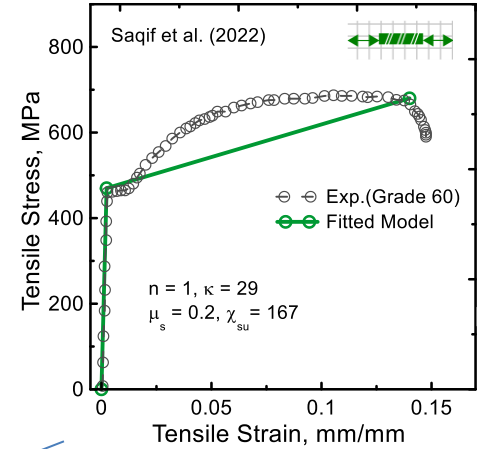
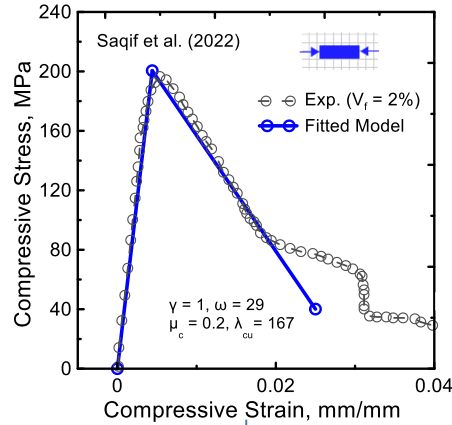
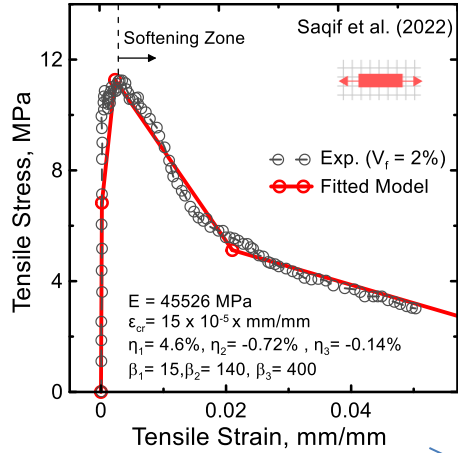
$$C_7 = \frac{\eta_1}{3} + ((\alpha^2 - 2\alpha + 1)\rho_c + \rho_t \alpha^2)n$$

$$C_8 = \frac{(k-1)^3 (\eta_1 - 1) \beta^2}{2} - \frac{(k-1)^3 (\eta_1 - 1)}{6}$$

Analytical Modeling

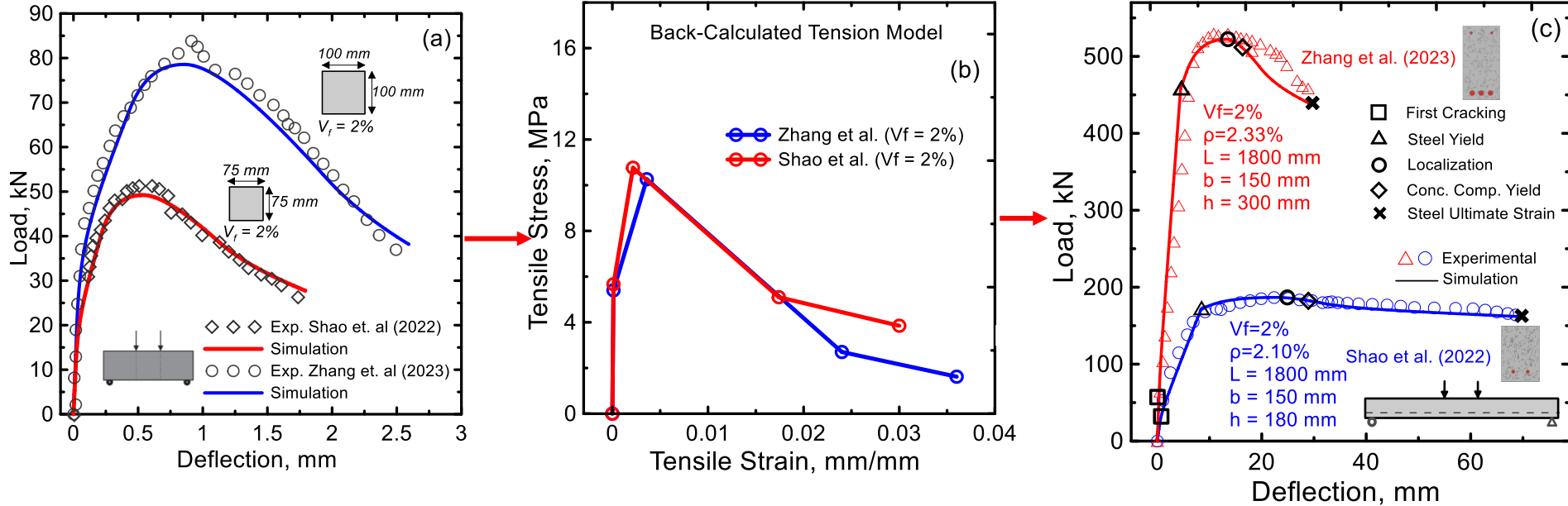


Flexure Modeling – Forward Analysis

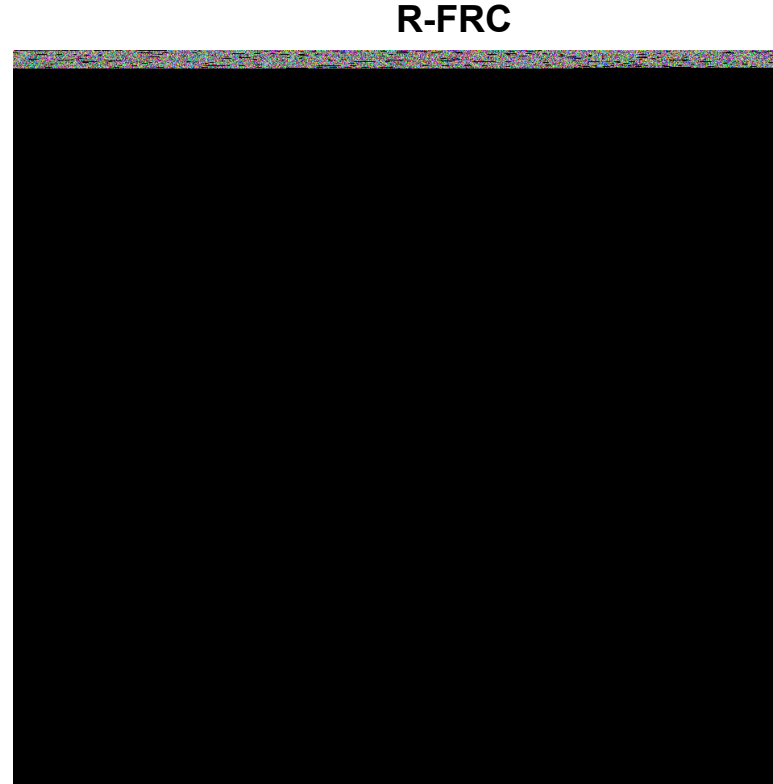
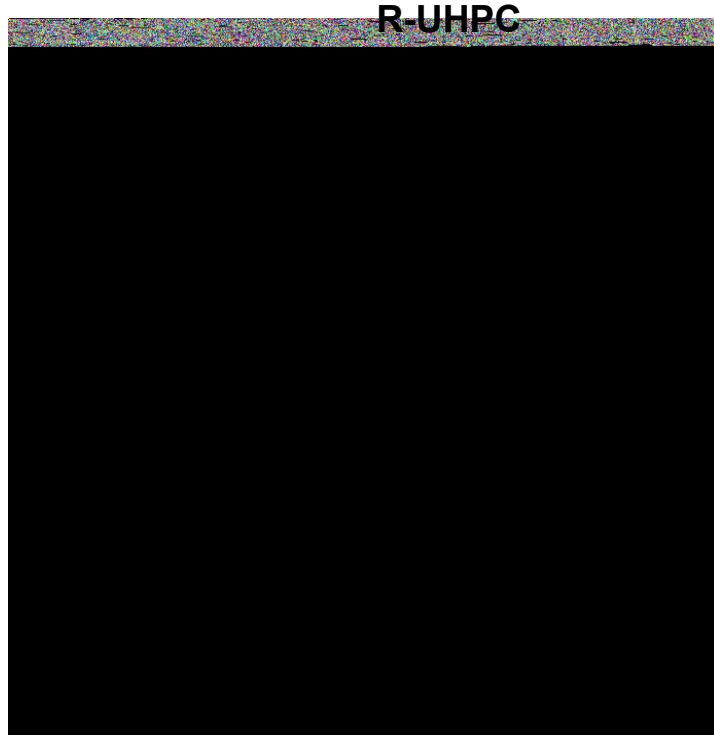


Predicted

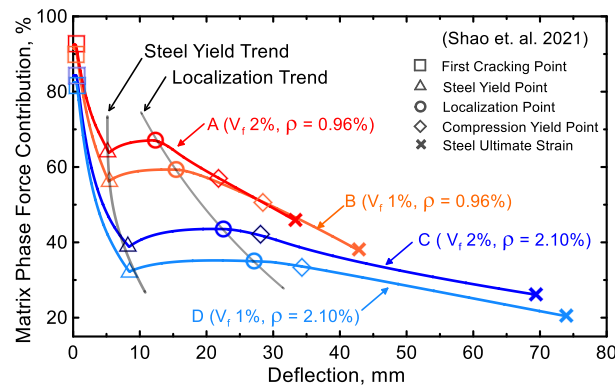
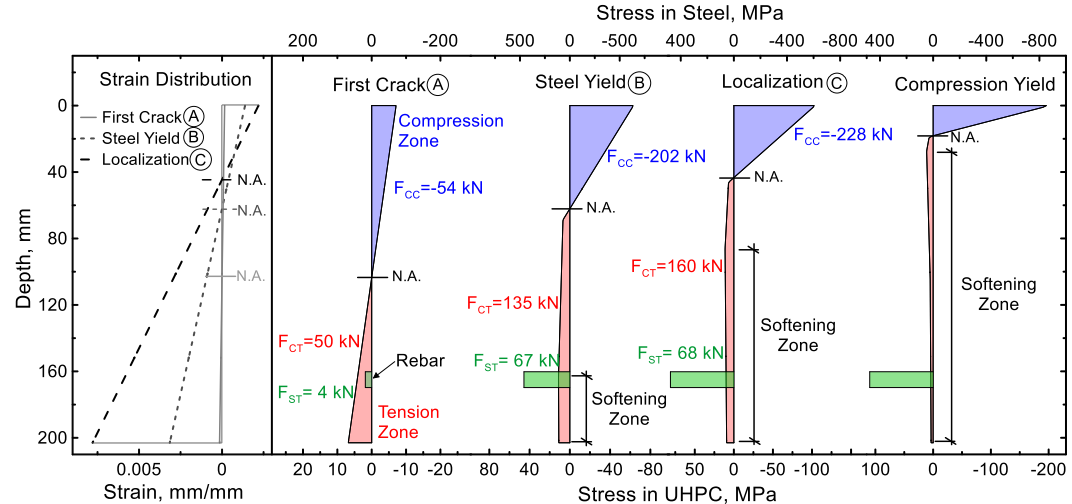
Flexure Modeling – Inverse Analysis



Flexure Modeling – Effect of Fibers of Hybrid Sections



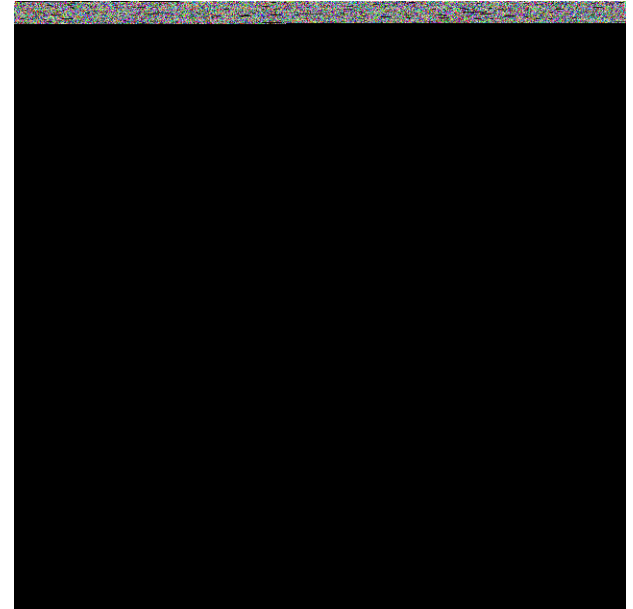
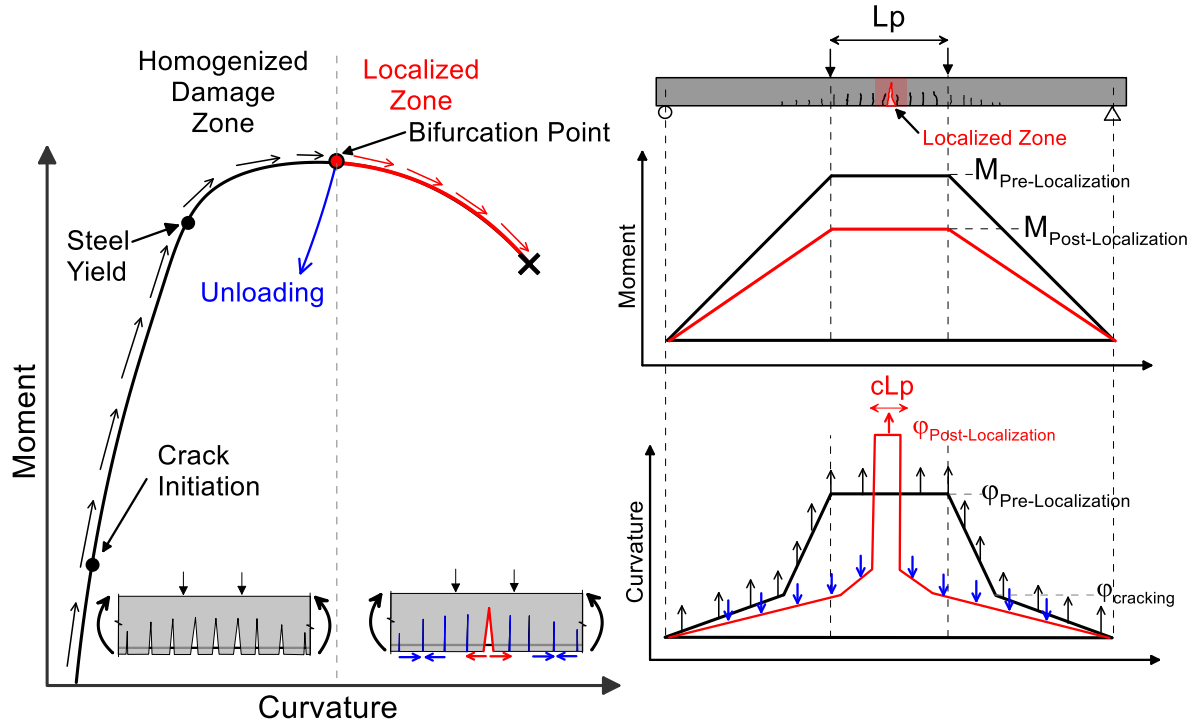
Strength Ductility Synergy in R-UHPC



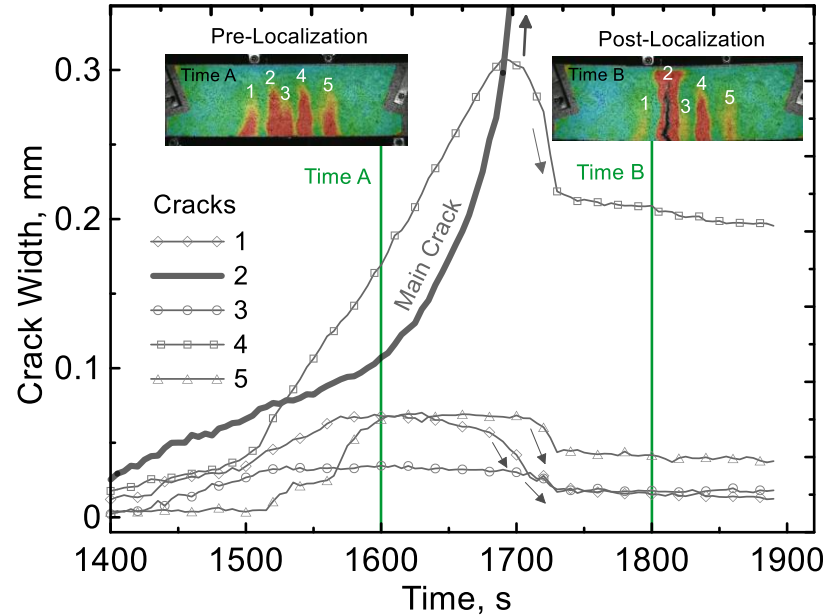
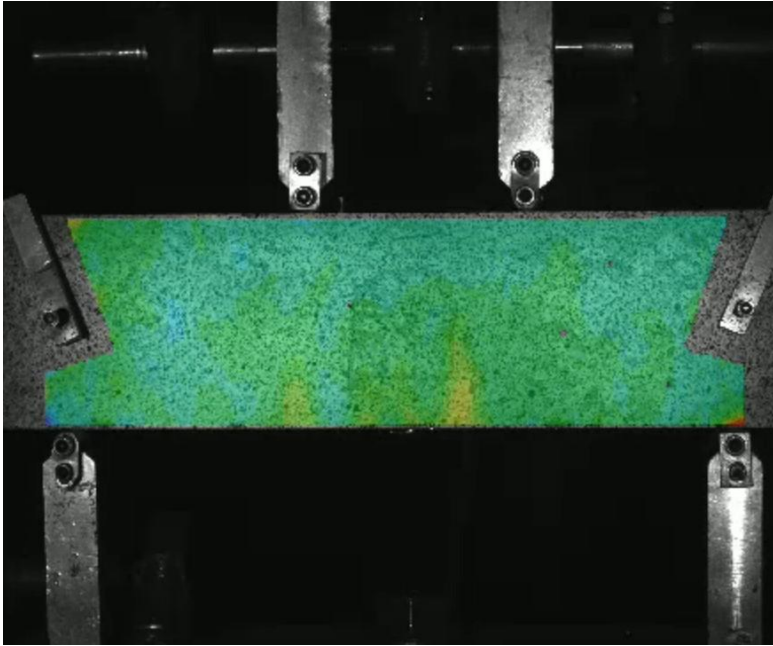
$$\psi_{UHP C} = \frac{F_{TC}}{F_{TS} + F_{TC}} = \frac{(n(\rho_c - \rho_t)\kappa + \omega)\mu}{n(\mu\rho_c + \omega\rho_t)\kappa + \mu\omega}$$

Flexure Modeling – Capturing Localization

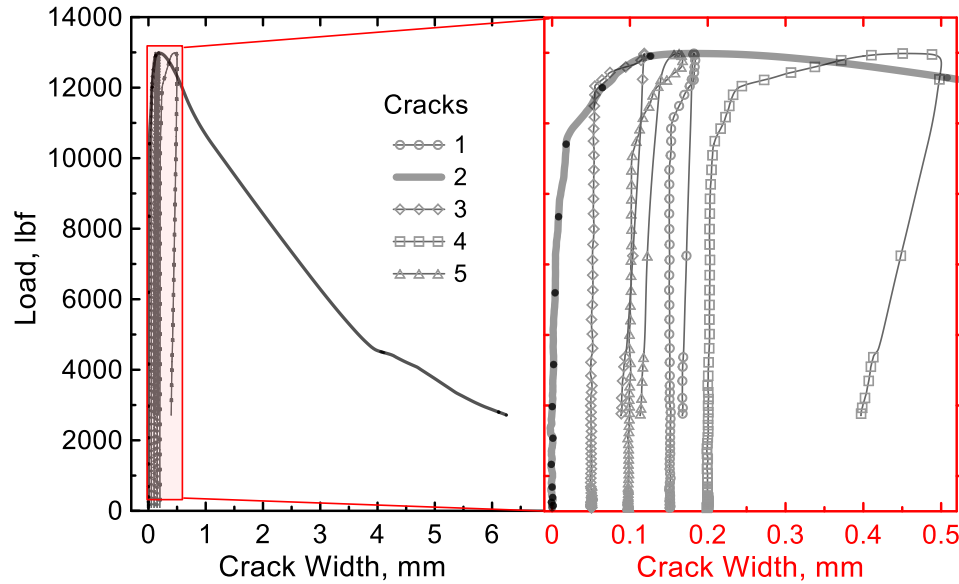
Addressing Plastic Length for Localization Induced ductility reduction in Reinforced UHPC



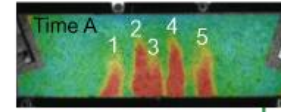
Flexure Modeling – Effect of Localization



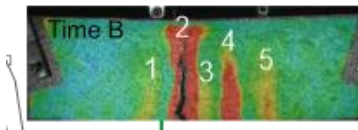
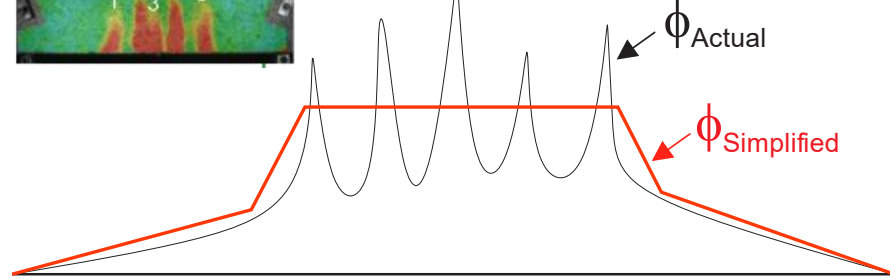
Flexure Modeling – Effect of Localization



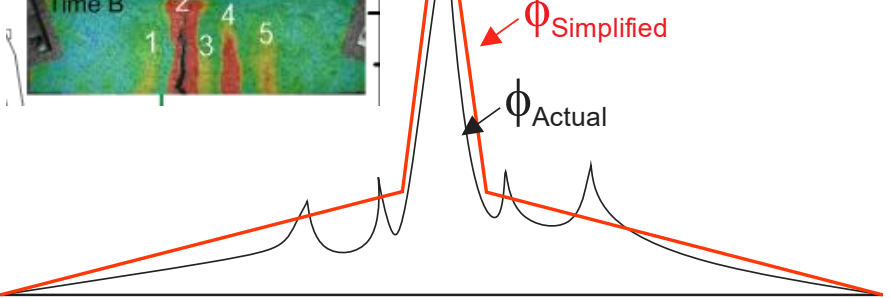
Curvature Distribution



Pre-Localization



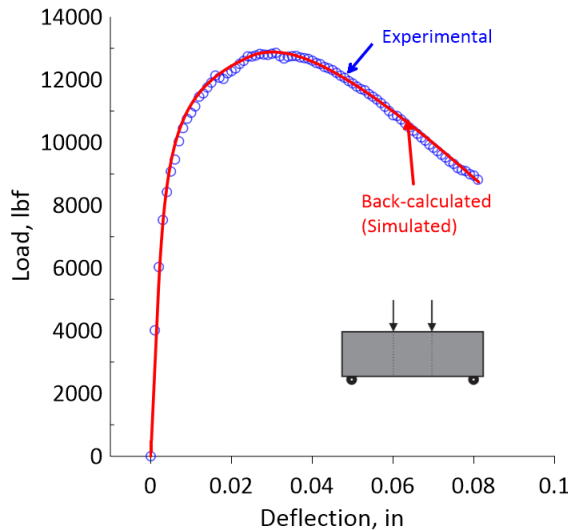
Post-Localization



ML Models for Inverse Analysis

- How to extract stress-strain from Flexural beam test?

Ans: Back-Calculation

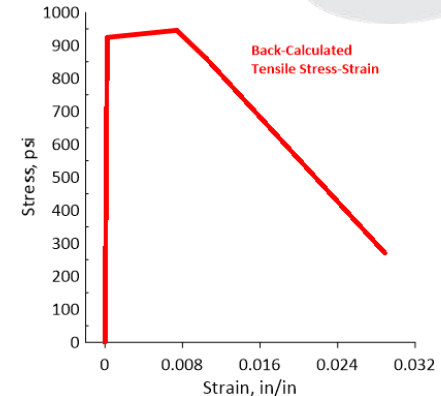
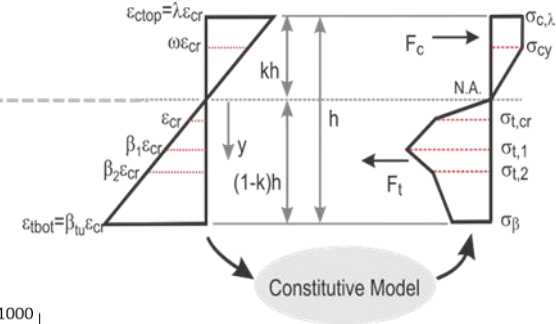


Goal is to Automate this using ML!

Start with an assumption of parameters for stress-strain

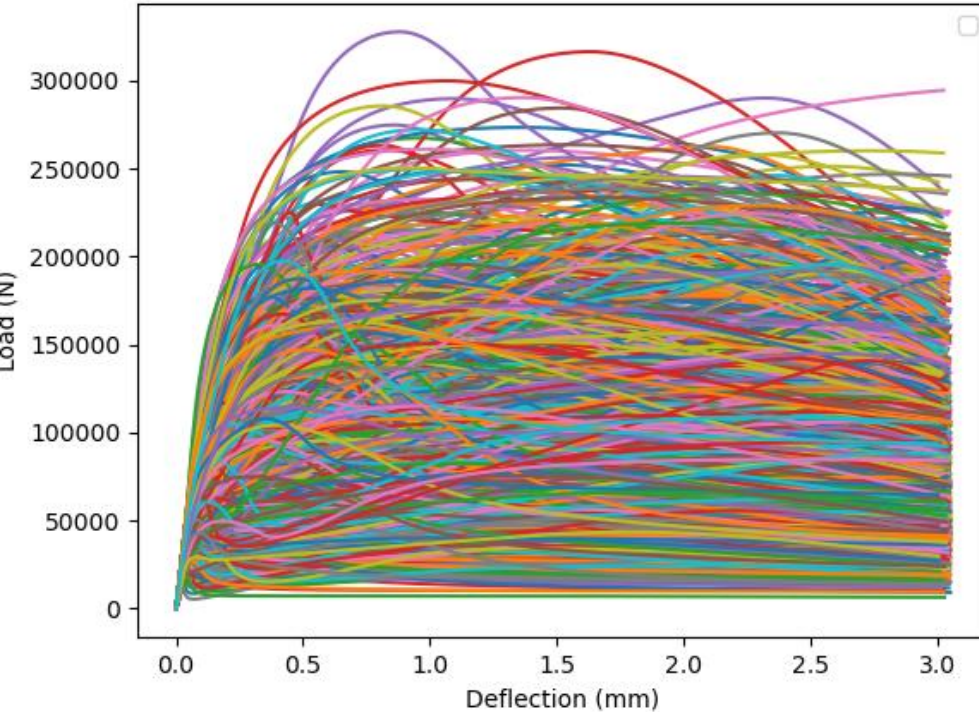
Fit Exp. Data

Get stress-strain

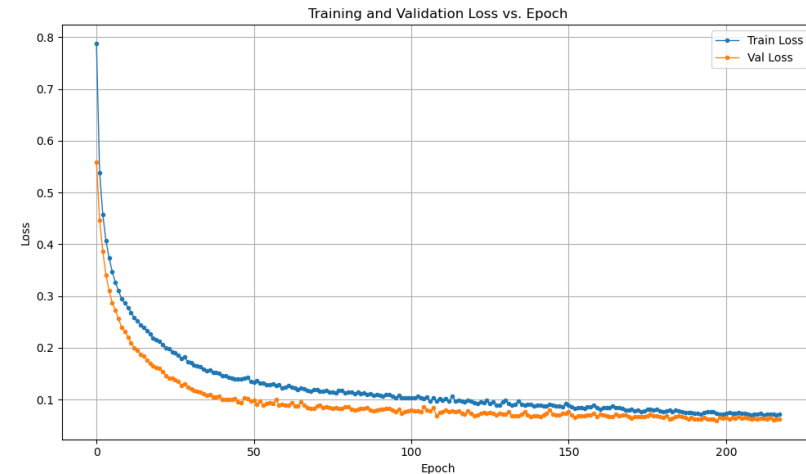


ML Models for Inverse Analysis

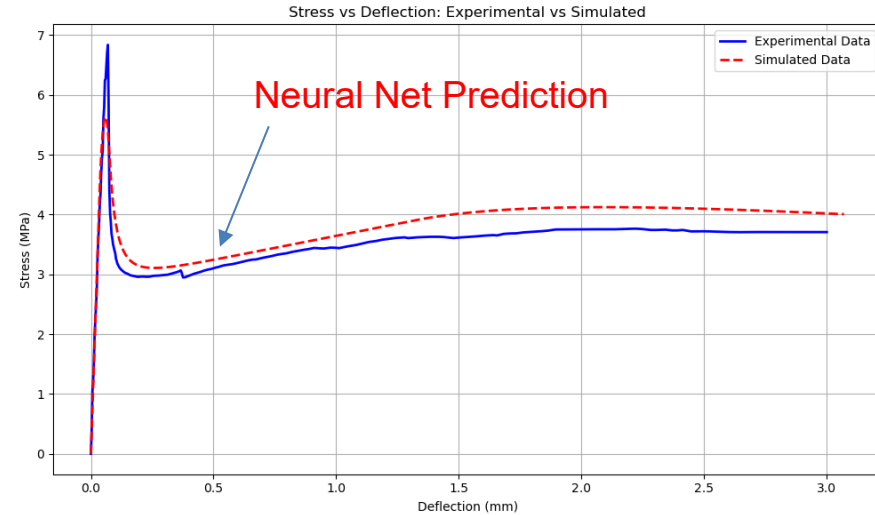
Large Experimental Data is required to train DNN for predicting tension model from Flexure Test.



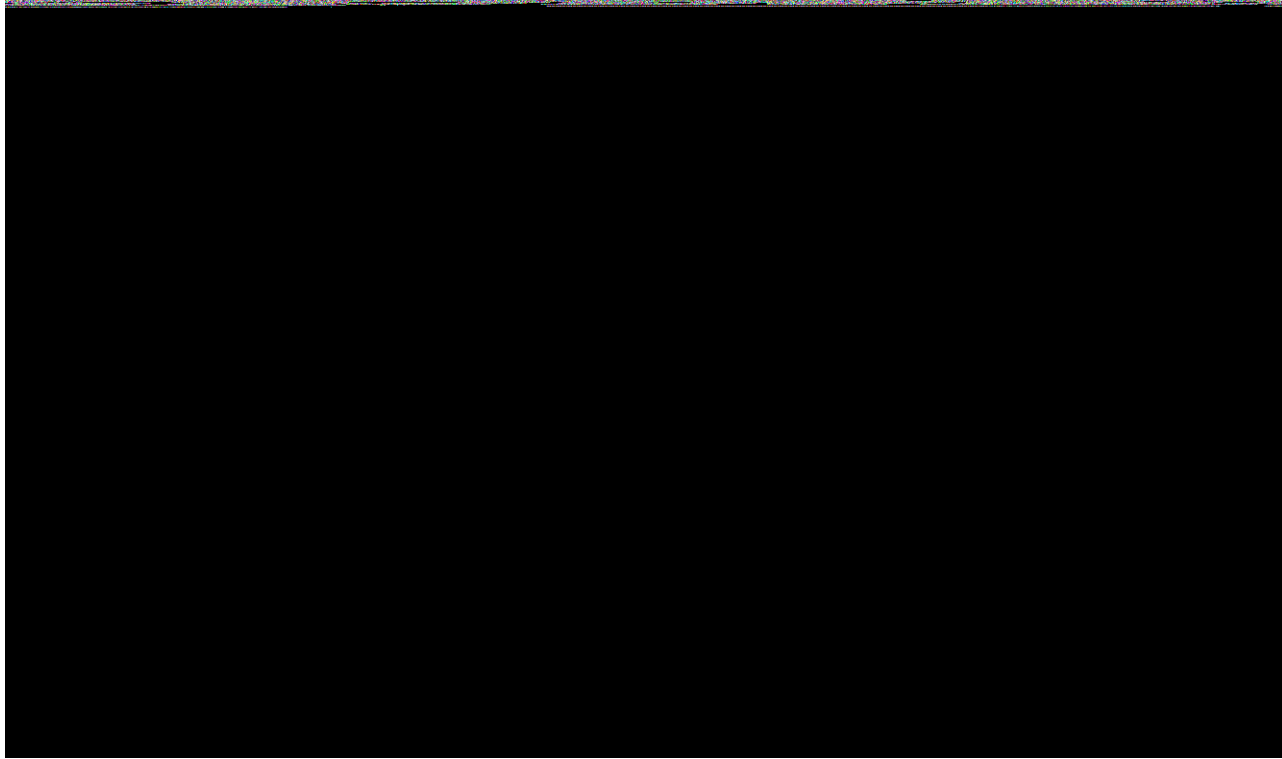
Analytical Closed-Form Models can generate 100,000 Load-deflection curves in 100 seconds on local PC as synthetic data for training Deep Neural Net.



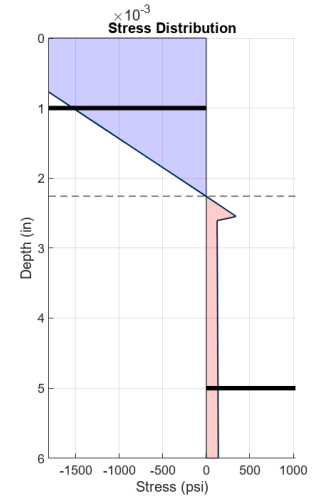
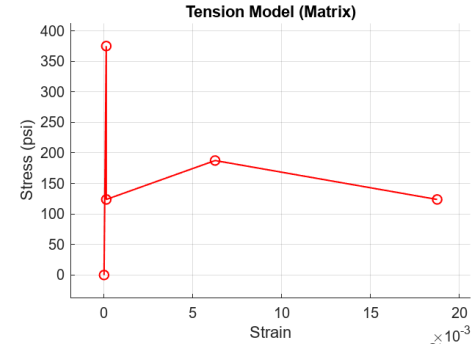
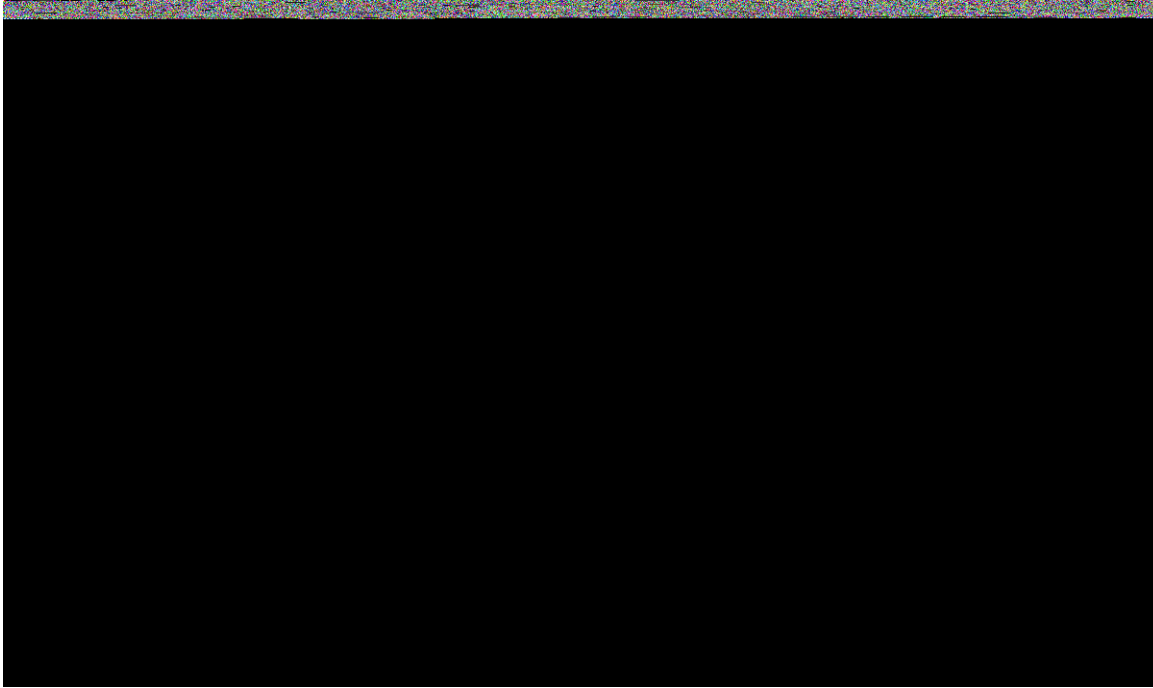
ML Models for Inverse Analysis



Software Applications



Software Applications



Inverse Analysis

This software provides a closed-form analytical platform for simulating, designing, and characterizing the flexural behavior of FRC, UHPC, and hybrid reinforced concrete systems. It supports inverse analysis of standard beam tests (ASTM C1609, EN 14651), design of custom reinforcement layouts, and generation of output plots and reports for research and engineering applications. The software is available for download on this link: [Software Download](#)

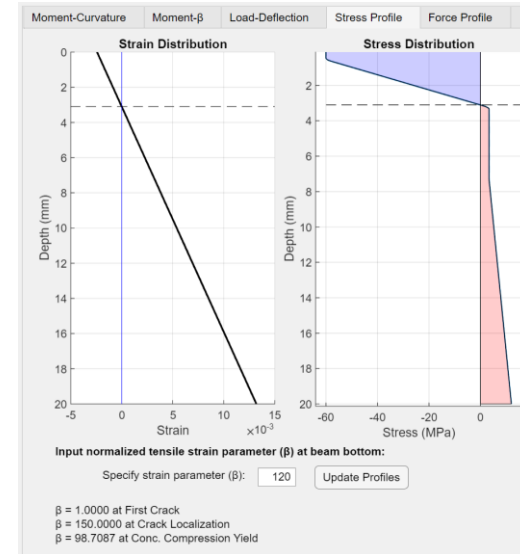


HRC DESIGNER & INVERSE ANALYSIS VERSION 3.4 (BETA)

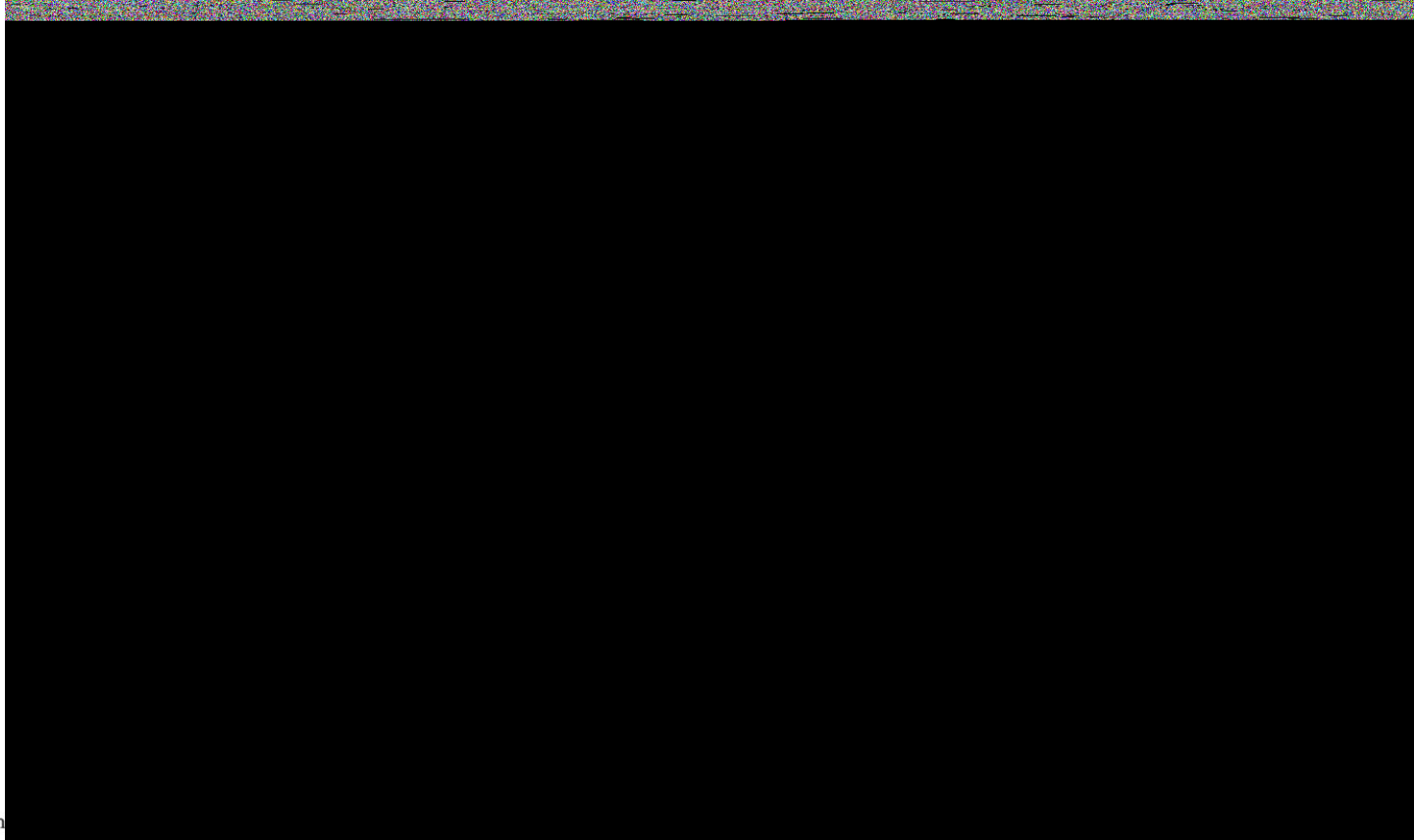
User Guide

03/2025

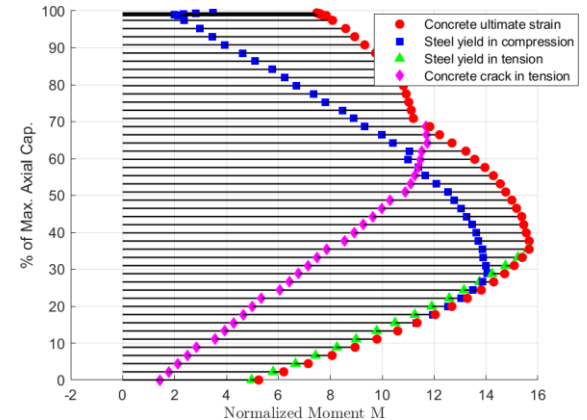
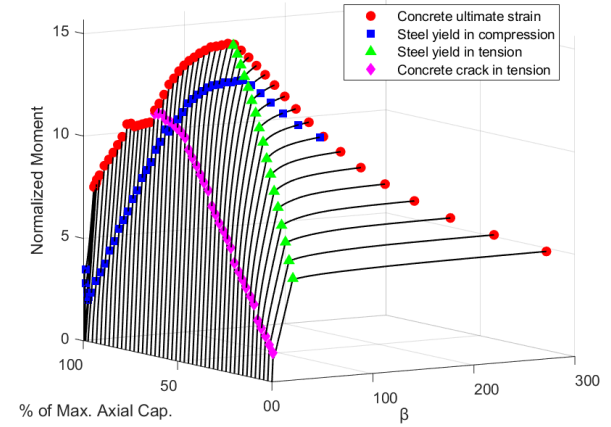
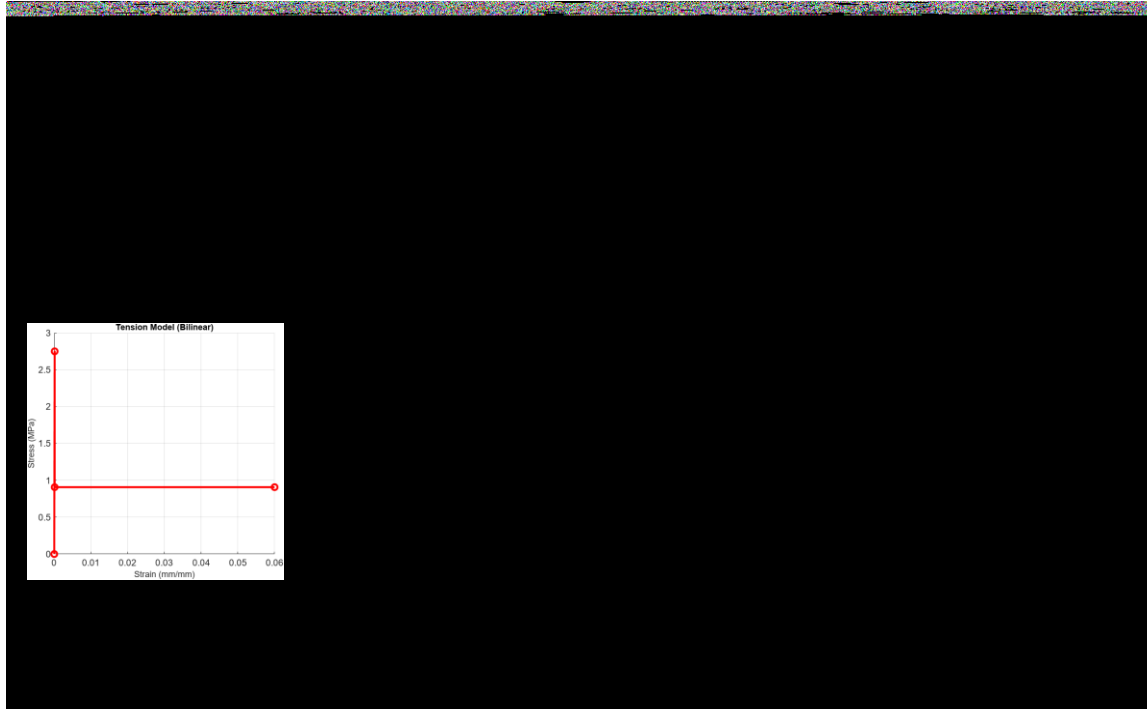
TRC Module (Forward and Inverse Modeling)



TRC Flexure Module (Forward and Inverse Modeling)

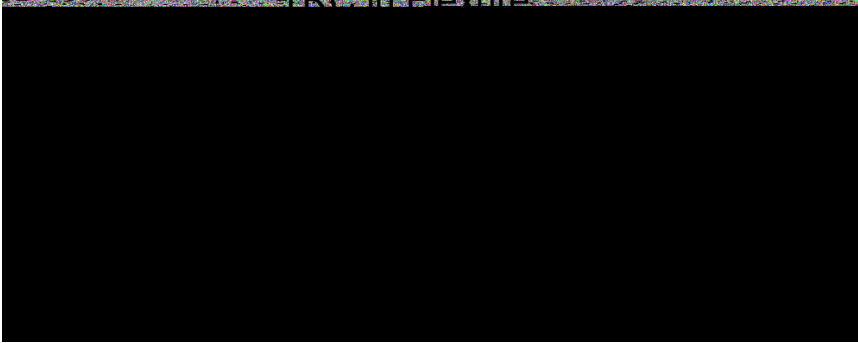


Column Design and P-M-Phi Diagrams

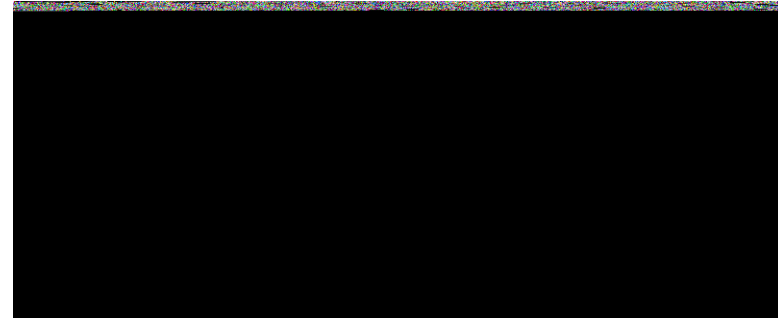
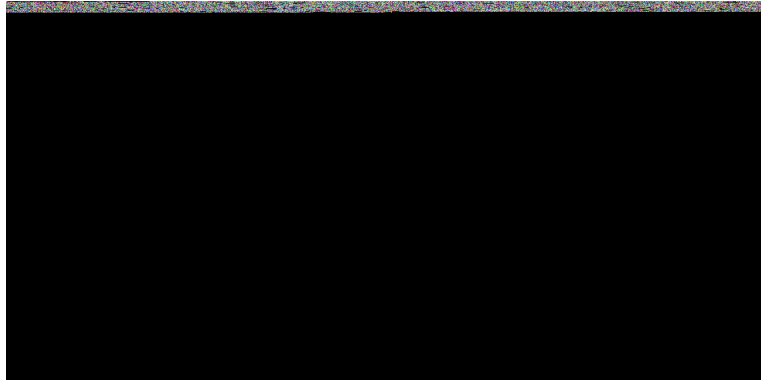
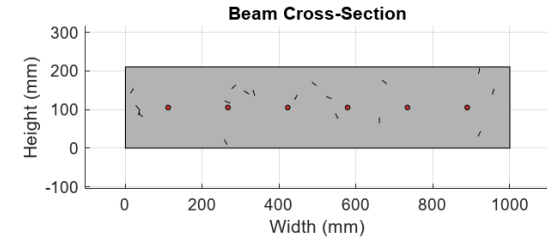


Other Modules

TRC in Flexure

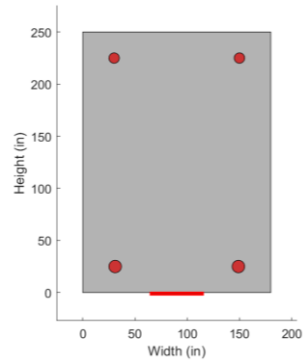


Slab on Grade

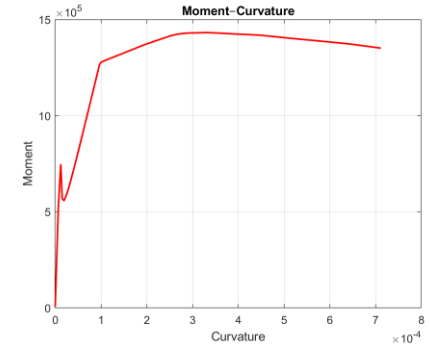


Other Modules

Strengthening Model



Iterative Models



Thank you

Questions?